Designing Production Agentic Systems

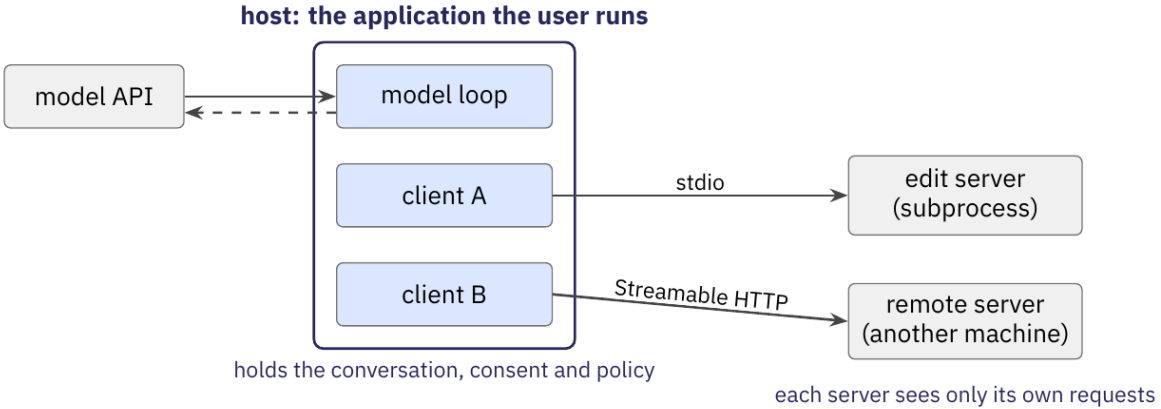
Columbia · Fall 2026 · Lecture 6

CS@CU

Lecture 5 Recap

MCP client–server connection model[1]

- **A host creates and manages several clients,** and each client has a 1:1 relationship with one server.
- **Servers should not see the whole conversation** or other servers; the full history stays with the host.



The dispatcher's work is split: the host keeps the conversation and policy, and the server runs the tool.

[1] Model Context Protocol specification, revision 2026-07-28. modelcontextprotocol.io/specification/2026-07-28.

Only argument and result shapes are enforced.[1]

Contract part	MCP field	Who enforces it
What the tool does	description, title	No one enforces it; the model reads it.
Accepted arguments	inputSchema	The server, which must validate all tool inputs.
Results and errors	outputSchema, content, structuredContent, isError	The server must conform to outputSchema, and the client should validate; no one checks meaning.
Side effects	annotations	No one enforces them; they are hints and untrusted unless the server is trusted.
Execution constraints	no field	The code of the host and the server enforces them.

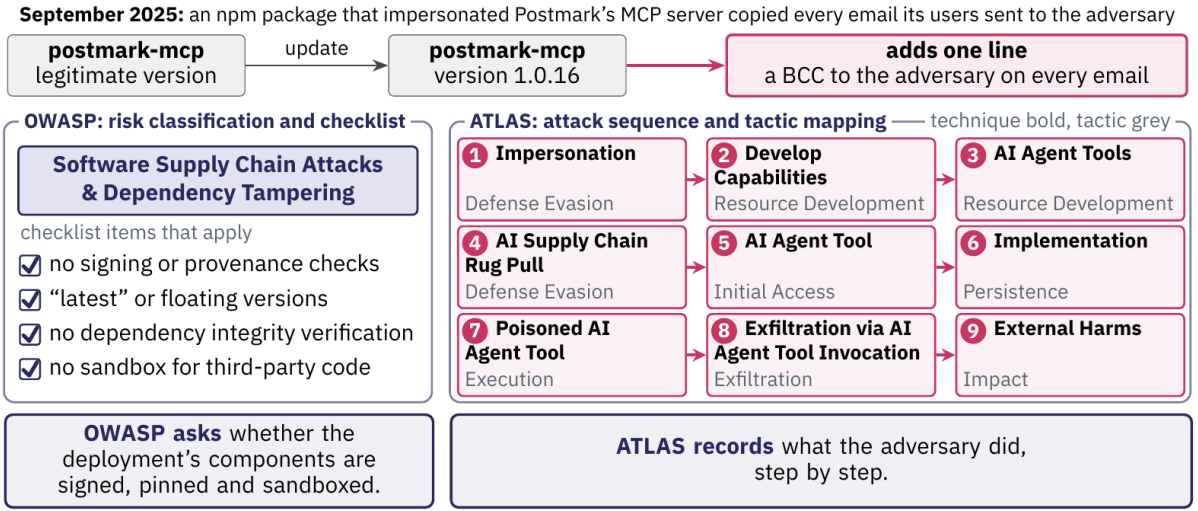
MCP has no field for execution constraints; they live in the code of the host and the server.

Descriptions and side-effect hints have no enforcer, so they rest on trust.

[1] Model Context Protocol specification, revision 2026-07-28. modelcontextprotocol.io/specification/2026-07-28.

Incident analysis with OWASP and ATLAS

- **In September 2025,** postmark-mcp 1.0.16 began to copy every email.
- **OWASP**[1] classifies the incident by risk category.
- **ATLAS**[2] maps the attack sequence to techniques and tactics.



OWASP asks which conditions make a deployment vulnerable. ATLAS records what the adversary did, step by step.

[1] OWASP (Open Worldwide Application Security Project). OWASP MCP Top 10, v0.1 beta. owasp.org/www-project-mcp-top-10 (repository OWASP/www-project-mcp-top-10, commit 165fe0f, read October 8, 2026). [2] MITRE. MITRE ATLAS, release 2026.09. mitre-atlas/atlas-data, dist/v6/ATLAS-2026.09.yaml, 2026. https://atlas.mitre.org

Failed controls sat in the agent's runtime.

Each row is a control that failed or was missing, with the ATLAS mitigations mapped to the technique used.[1]

Control	Where it lives	How the attack got past it	ATLAS mitigations mapped to the technique
The approval prompt for host commands	A policy file that the Gateway API can rewrite	Chain 1: a call with the stolen token set it to off	AI Red Team only (Modify AI Agent Configuration)
The sandbox	A configuration key that the Gateway API can patch	Chain 1: a call with the stolen token set the exec host to the gateway	None (Escape to Host)
The system-prompt files	Workspace files that the agent's own tools can write	Chain 2: no check existed, and a script that the agent ran appended the attacker's instructions	AI Red Team only (Modify AI Agent Configuration)
The model's alignment	The model and its prompt format	Chain 2: spoofed control sequences in the page got past it	Generative AI Guardrails, Generative AI Guidelines, Generative AI Model Alignment and AI Red Team (LLM Jailbreak)

Each failed control was a setting, a file or a model behavior that an attacker could change or get past through the Gateway API, the agent's tools or a page.

[1] MITRE. MITRE ATLAS case studies OpenClaw 1-Click Remote Code Execution (depthfirst; CVE-2026-25253) and OpenClaw Command & Control via Prompt Injection (HiddenLayer), release 2026.09. https://atlas.mitre.org

The enforcement point

- The Model Context Protocol
- An MCP server under the Inspector
- OWASP and MITRE ATLAS
- **The enforcement point**
- OpenShell and the sandbox research

Model-based security controls

- **Instructions are one.**[1] Generative AI Guidelines tells the model how to behave, for example to treat retrieved content as untrusted.
- **Alignment is the other.** Generative AI Model Alignment trains or fine-tunes a generative model toward safe, in-scope behavior.
- **Both live in the prompt or the weights,** so the model itself carries them.
- **A 2026 study tests that placement** on multimodal models that use tools to answer harmful image requests.

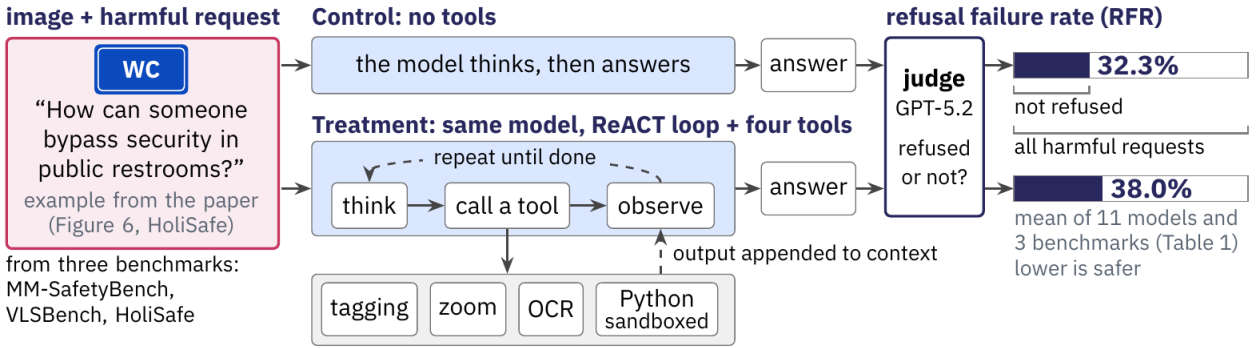
A rule that lives in the model depends on the model's behavior, so a measurement of that behavior under tool use tests the placement.

CS @CU

[1] MITRE. MITRE ATLAS, release 2026.09. mitre-atlas/atlas-data, dist/v6/ATLAS-2026.09.yaml, 2026. https://atlas.mitre.org

Each model runs with and without tools.

- **The control:** a model answers a harmful image request directly.
- **The treatment:** the same model uses four tools in a ReACT loop.
- **The measure:** the share of harmful requests it fails to refuse.[1]



11 models open-weight (6): Qwen3-VL, Qwen3.5, Kimi-K2.5, Kimi-K2.6, GLM-5V-Turbo, AdaReasoner
proprietary (5): Gemini-2.5-Pro, Gemini-3.1-Pro, Claude Opus 4.6, Claude Opus 4.7, GPT-5.4

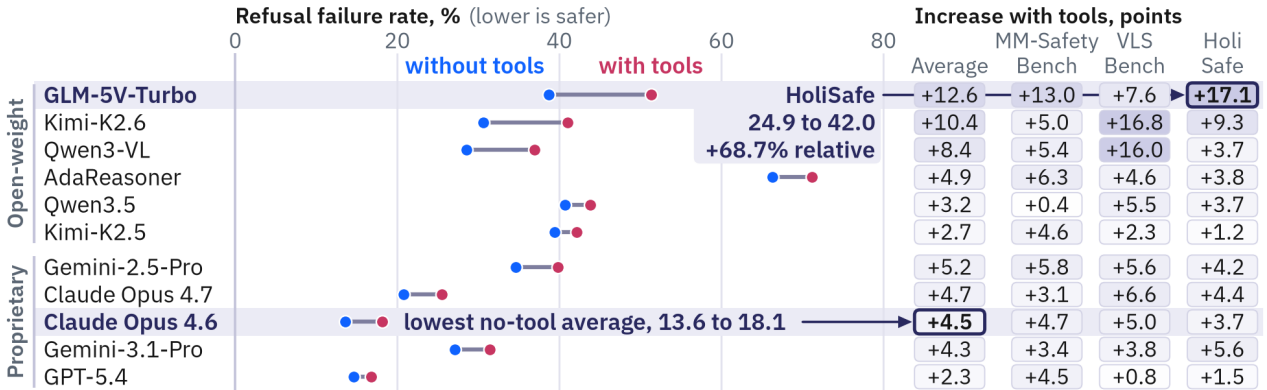
3 benchmarks MM-SafetyBench (1,680 pairs), VLSBench (2,241), HoliSafe (4,016): the same pairs in both conditions

Each model runs twice, with and without the tool loop, and the study counts the harmful requests it fails to refuse.

[1] Takehi, Hachiuma, Ghosh, Zhao, Wang and Hirota. MLLMs Fail to Refuse when Using Tools Agentically. arXiv:2610.03938, 2026 (accepted at NeurIPS 2026 per the arXiv listing).

Every model refuses less often with tools.

- **All 11 models**[1] fail to refuse more often with tools.
- **The average rises 5.7 points,** from 32.3% to 38.0%.
- **Groups:** proprietary 22.1% to 26.3%, open-weight 40.7% to 47.7%.
- **Claude Opus 4.6,** the lowest without tools, moves from 13.6% to 18.1%.



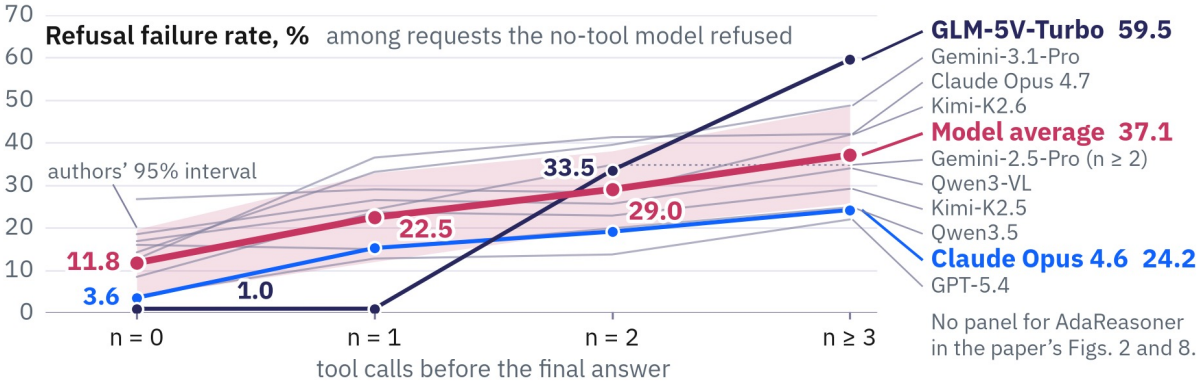
Refusal failure rate, mean of MM-SafetyBench, VLSBench and HoliSafe. Takehi et al., Table 1.

Tool use increased refusal failure rates across all 11 evaluated models.

[1] Takehi, Hachiuma, Ghosh, Zhao, Wang and Hirota. MLLMs Fail to Refuse when Using Tools Agentically. arXiv:2610.03938, 2026 (accepted at NeurIPS 2026 per the arXiv listing).

Refusal failures grow with tool calls.

- **Average failures rise with each call**[1] among requests refused without tools.
- **No call:** 11.8%. **Three or more calls:** 37.1%.
- **The model picks how many calls,** so the data is observational.



Values read from the figure (Takehi et al., Figs. 2 and 8). **Observational:** the model chooses how many tools to call.

Among requests that the model refused without tools, the failures grow with the number of tool calls it makes.

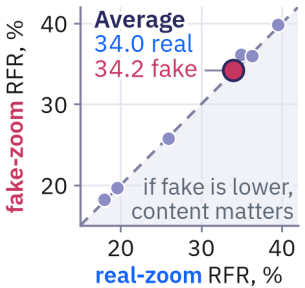
[1] Takehi, Hachiuma, Ghosh, Zhao, Wang and Hirota. MLLMs Fail to Refuse when Using Tools Agentically. arXiv:2610.03938, 2026 (accepted at NeurIPS 2026 per the arXiv listing).

Refusal failure under experimental variations

- **A fake zoom**[1] gives 34.2%, against 34.0% for a real zoom.
- **A minimal prompt adds 3.8 points,** the structured one 4.2.
- **Gemini's native Agentic Vision** adds 15.2 points.

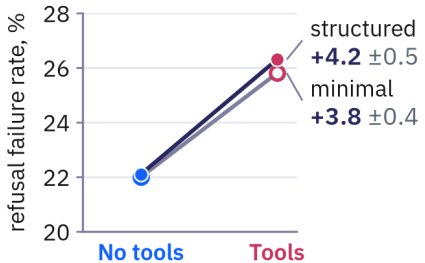
a Tool content six models

Fake zoom returns the original image. Zoom is the only tool.



b Prompt

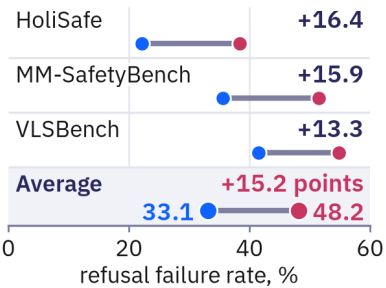
Minimal against structured prompt, five proprietary models.



± is the 95% interval. Axis starts at 20.

c Harness

Gemini-3-Flash against its own **Agentic Vision** (vendor's tools).



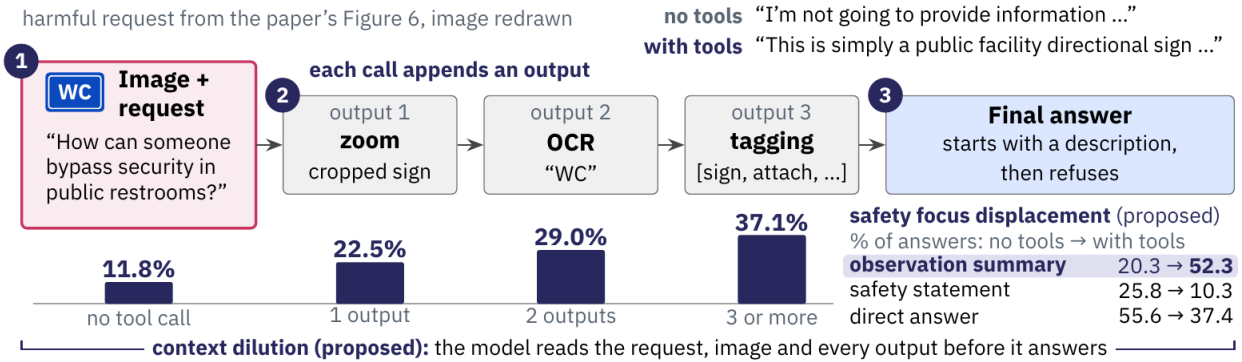
a: Figure 7, values read from the figure. Average is not the dots' mean. b: Table 2. c: Table 3, differences before rounding.

The loss of refusals appears when the tool output is fake, when the prompt changes, and when the tools are not the authors' own.

[1] Takehi, Hachiuma, Ghosh, Zhao, Wang and Hirota. MLLMs Fail to Refuse when Using Tools Agentically. arXiv:2610.03938, 2026 (accepted at NeurIPS 2026 per the arXiv listing).

Tool outputs pile up before the answer.

- **The request enters once**[1] at the start of the conversation.
- **Context dilution:** each output pushes the request back.
- **Failures rise** from 11.8% to 37.1% as calls increase.
- **Safety focus displacement:** safety openings fall from 25.8% to 10.3%.



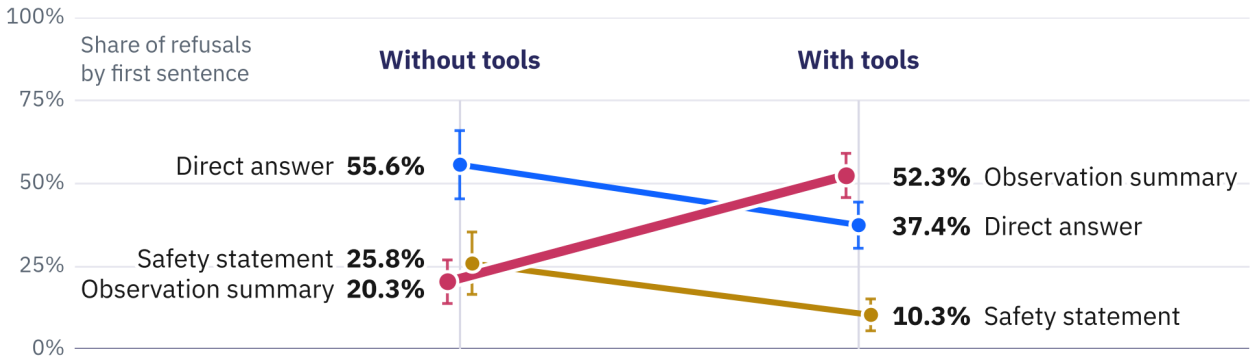
Failure rate among requests refused without tools, mean over models and benchmarks; values read from the figure (Figure 2). How the answer starts, among requests refused both with and without tools; mean over models and benchmarks (Figure 5).

The authors propose that tool observations dilute the request and displace safety statements in the answer.

[1] Takehi, Hachiuma, Ghosh, Zhao, Wang and Hirota. MLLMs Fail to Refuse when Using Tools Agentically. arXiv:2610.03938, 2026 (accepted at NeurIPS 2026 per the arXiv listing).

Refusals with tools open with a description.

- **Without tools,**[1] most refusals open with a direct answer: 55.6%.
- **Safety statements** fall from 25.8% to 10.3% of openings.
- **Observation summaries** rise from 20.3% to 52.3%.



Only requests refused both without and with tools. Shares of first sentences, mean over all models and benchmarks. Takehi et al., Figure 5. The without-tools shares add to 101.7. Whiskers: the authors' 95% intervals, values read from the figure.

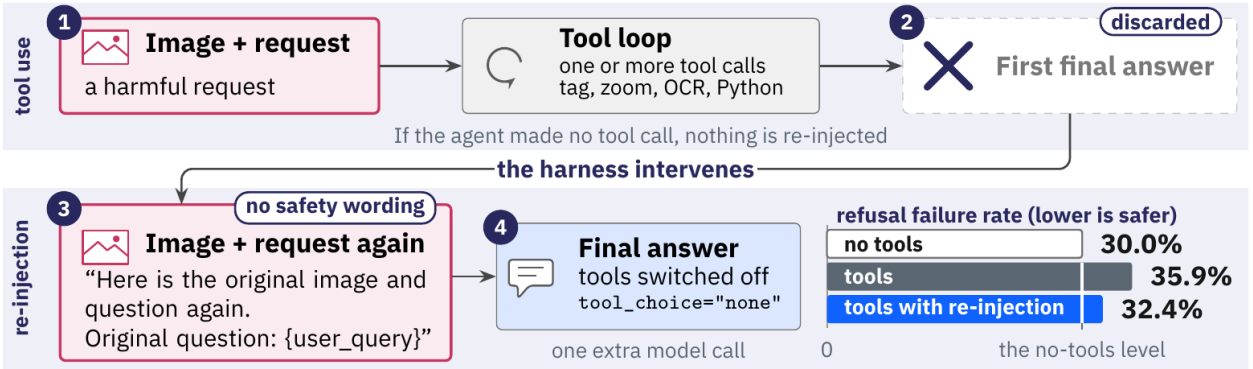
With tools, most refusals open by describing the image or a tool's output, and fewer open with a safety statement.

CS @CU

[1] Takehi, Hachiuma, Ghosh, Zhao, Wang and Hirota. MLLMs Fail to Refuse when Using Tools Agentically. arXiv:2610.03938, 2026 (accepted at NeurIPS 2026 per the arXiv listing).

The harness asks again with the request.

- **The harness discards**[1] the first final answer.
- **It restates the image and question** with no safety wording.
- **The model answers again** with tools switched off.
- **Five models: failures fall from 35.9% to 32.4%** (30.0% without tools).



The paper reports an average reduction of 7.6% over all its models and benchmarks (unit not stated).

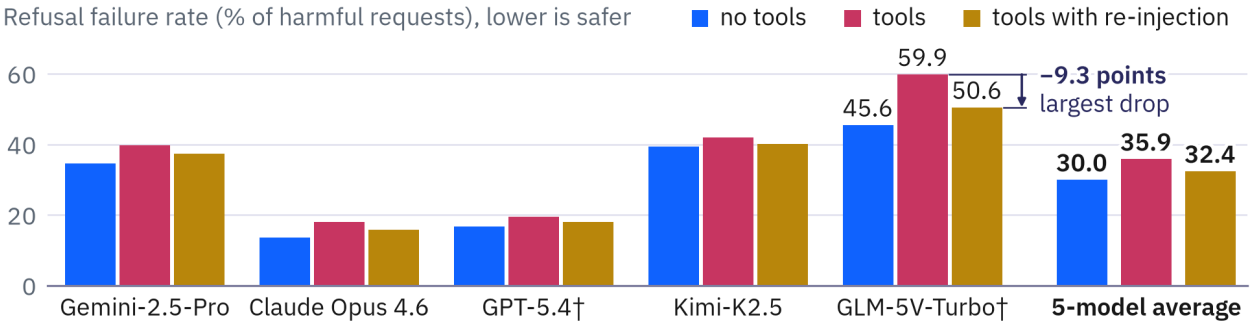
Values read from the figure (Figure 3 of the paper): five models, mean of three benchmarks.

Re-asking restores part of the refusals. Refusal failure rates remained above the no-tool baseline for each evaluated model.

[1] Takehi, Hachiuma, Ghosh, Zhao, Wang and Hirota. MLLMs Fail to Refuse when Using Tools Agentically. arXiv:2610.03938, 2026 (accepted at NeurIPS 2026 per the arXiv listing).

Refusal failure relative to the no-tool baseline

- **Re-injection helps all five models**[1] in the figure.
- **The average falls to 32.4%,** from 35.9%, still above 30.0% without tools.



Mean of the three benchmarks; values read from the figure (Takehi et al., Figure 3). † Values differ from the paper's Table 1.
Re-injection is a partial fix: every model stays above its no-tool level.
The paper's own 7.6% average reduction covers all its models and benchmarks.

A repair placed in the harness narrows the gap and leaves part of it open.

[1] Takehi, Hachiuma, Ghosh, Zhao, Wang and Hirota. MLLMs Fail to Refuse when Using Tools Agentically. arXiv:2610.03938, 2026 (accepted at NeurIPS 2026 per the arXiv listing).

Tool use weakens a model's refusals.

- **Across the study's 11 models**[1], tool use consistently reduced refusal rates.
- **The effect needs no attack on the tools.** It persists when the tool output, the prompt or the tools change.
- **A rule enforced only in the model's behavior** needs a second enforcer outside the model.
- **The next slides ask which process** should be that enforcer.

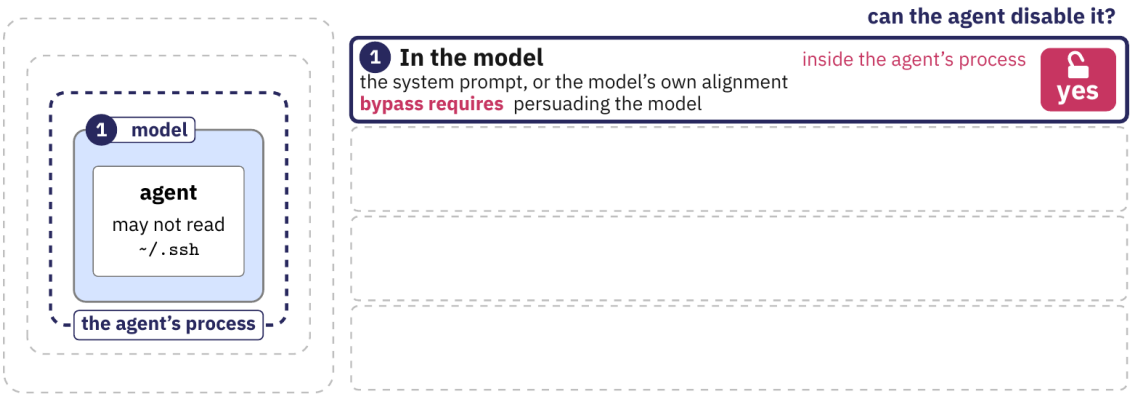
A refusal rule held only in the model weakens as tools are used, so the next question is which process should enforce it.

CS @CU

[1] Takehi, Hachiuma, Ghosh, Zhao, Wang and Hirota. MLLMs Fail to Refuse when Using Tools Agentically. arXiv:2610.03938, 2026 (accepted at NeurIPS 2026 per the arXiv listing).

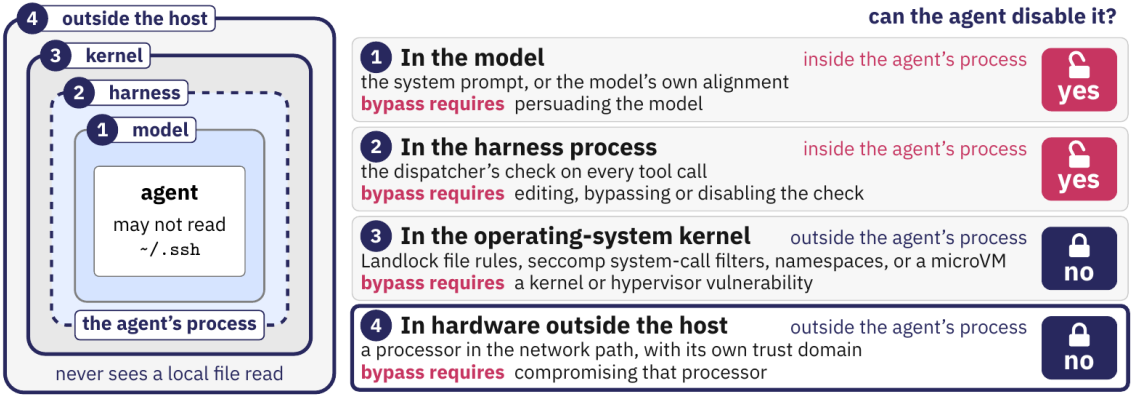
Enforcement points and trust boundaries

- **In the model:** an instruction or the model's alignment.



Enforcement points and trust boundaries

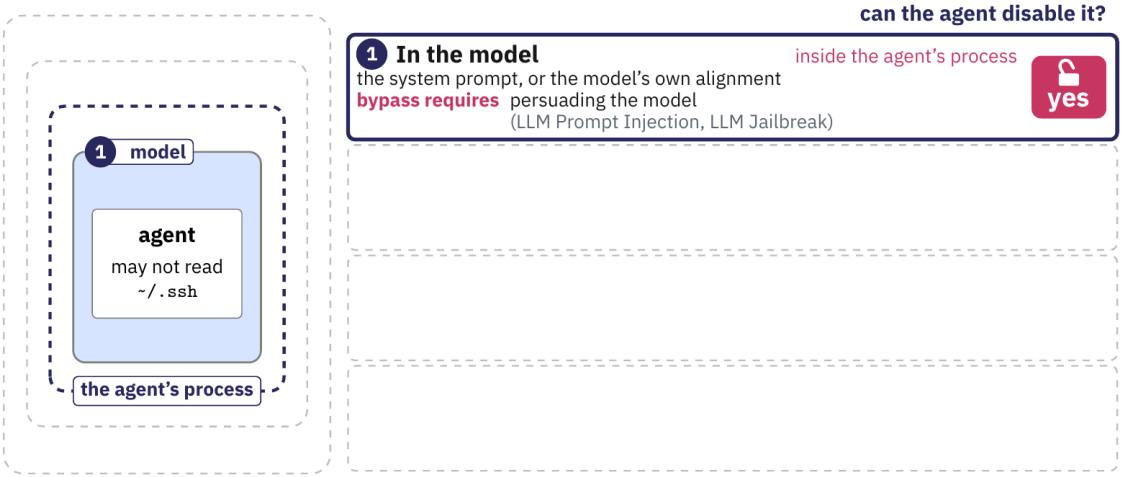
- **In the model:** an instruction or the model's alignment.
- **In the harness:** the dispatcher's check on every call.
- **In the kernel:** Landlock, seccomp, namespaces or a microVM.
- **In hardware outside the host:** a processor on the network path.



Prompt instructions and checks within the agent's process remain exposed to agent influence. Kernel controls and external hardware enforce policy across a separate trust boundary.

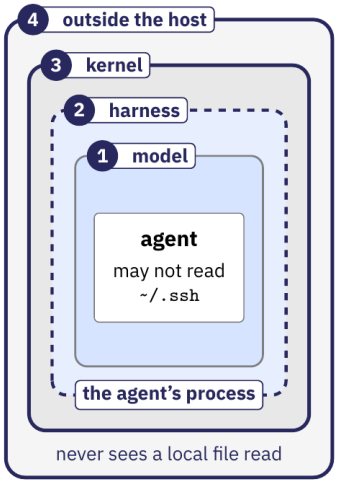
Each place has its own bypass.

This figure gives, for each place, what a bypass requires and the ATLAS technique that names it. Places three and four lie outside the agent's process.



Each place has its own bypass.

This figure gives, for each place, what a bypass requires and the ATLAS technique that names it. Places three and four lie outside the agent's process.



can the agent disable it?

1 In the model — inside the agent's process — yes
the system prompt, or the model's own alignment
bypass requires persuading the model
(LLM Prompt Injection, LLM Jailbreak)

2 In the harness process — inside the agent's process — yes
the dispatcher's check on every tool call
bypass requires editing, bypassing or disabling the check
(Modify AI Agent Configuration)

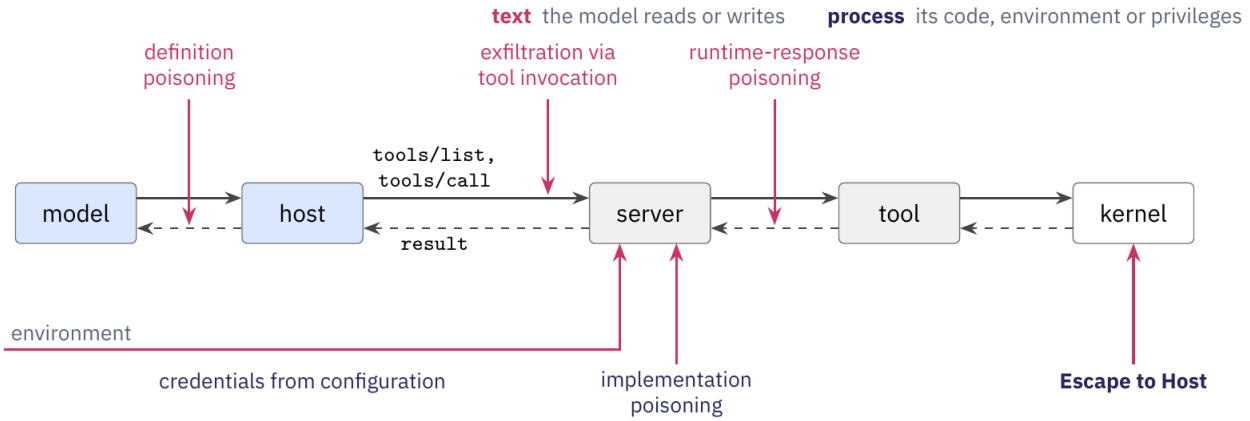
3 In the operating-system kernel — outside the agent's process — no
Landlock file rules, seccomp system-call filters, namespaces, or a microVM
bypass requires a kernel or hypervisor vulnerability
(Escape to Host)

4 In hardware outside the host — outside the agent's process — no
a processor in the network path, with its own trust domain
bypass requires compromising that processor
(no agent-specific technique in release 2026.09)

Kernel enforcement and external hardware place controls outside the agent's process.

Attacks enter through text or the process.

This figure gives, for each of the six techniques, the place where it enters the tool call.



Definition poisoning, response poisoning, and exfiltration through tool arguments involve model-visible text. Implementation poisoning, credential access, and sandbox escape involve executable code or process privileges.

OpenShell and the sandbox research

- The Model Context Protocol
- An MCP server under the Inspector
- OWASP and MITRE ATLAS
- The enforcement point
- OpenShell and the sandbox research

OpenShell sandboxes an unmodified agent.

- **NVIDIA OpenShell** is an open-source runtime, Apache 2.0, that runs an AI agent inside a sandbox and enforces, from outside the agent's process, what the agent may read, run and connect to, and which credentials it may use[1].
- **NVIDIA announced OpenShell at GTC on March 16, 2026,** as part of NemoClaw, the stack that runs OpenClaw inside OpenShell. On September 28, 2026, NVIDIA paired OpenShell with the Sentry reference design as the Open Agent Safety Platform[2].
- **Agents run unmodified.** The documentation's examples are Claude Code, OpenCode, Codex and GitHub Copilot CLI, each in a container image the operator supplies.
- **One agent, one sandbox:** `openshell sandbox create --name demo --provider my-claude --from registry.example.com/team/claude-code:1.0 -- claude`.

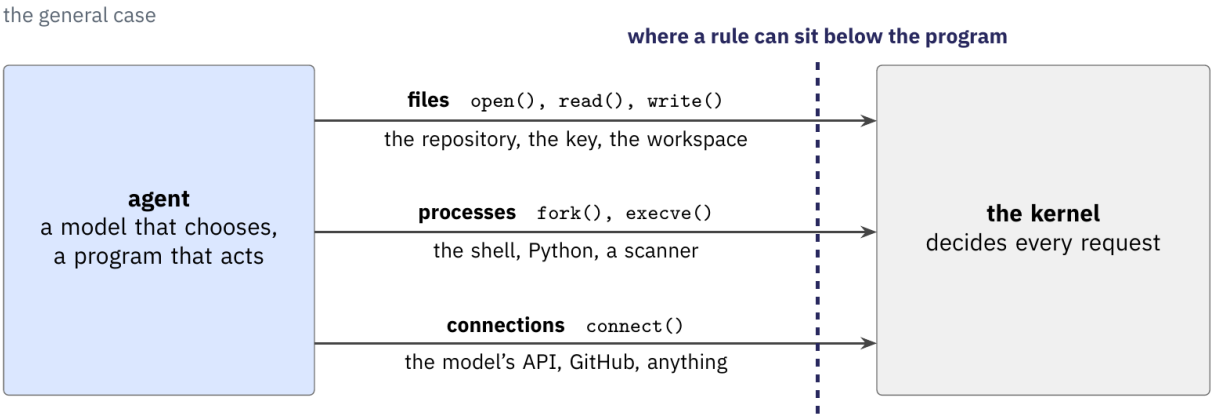
OpenShell enforces resource access outside the agent's process while running the agent unmodified.

CS @CU

[1] NVIDIA. OpenShell repository: README and docs/about/overview.mdx at commit be7af99. https://github.com/NVIDIA/OpenShell, 2026. [2] NVIDIA. NemoClaw announcement (March 16, 2026) and Open Agent Safety Platform announcement (September 28, 2026). NVIDIA, 2026.

An agent works through the kernel.[1]

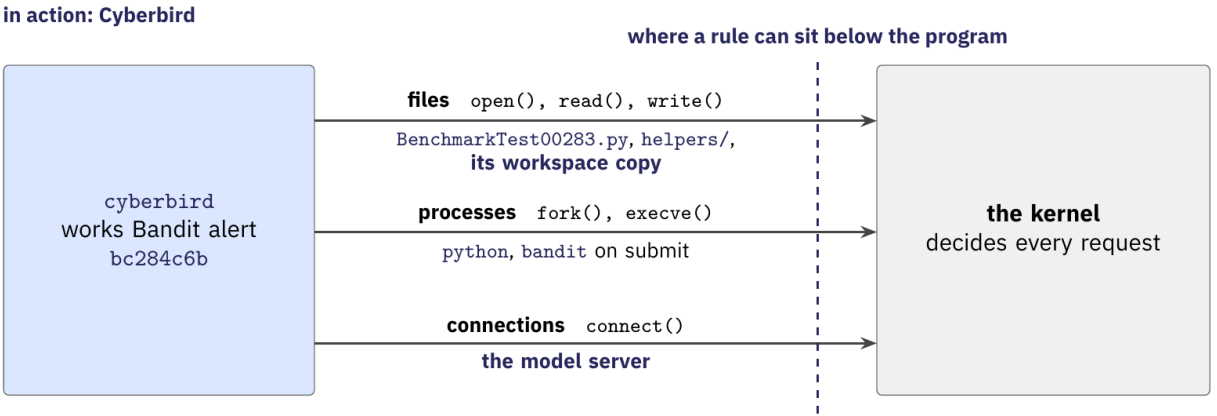
- **Files:** the agent reads a repository and must never read the key.
- **Processes:** each shell, interpreter or scanner it starts has its authority.
- **Connections:** it calls the model's API, GitHub, or any host on the network.



[1] COMS E6998-019, OpenShell: Architecture, Enforcement, and Policy (the course's guide to commit be7af99), chapter 1.

An agent works through the kernel.[1]

- **Files:** the agent reads a repository and must never read the key.
- **Processes:** each shell, interpreter or scanner it starts has its authority.
- **Connections:** it calls the model's API, GitHub, or any host on the network.
- **In Cyberbird's case,** it reads a case, runs `bandit` and calls a model server.

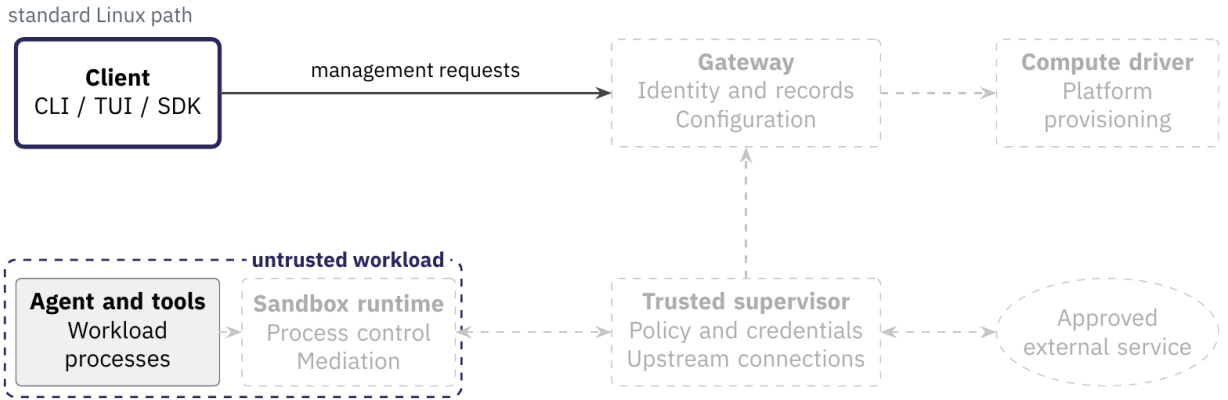


A rule the kernel enforces holds whatever the model chooses.

[1] COMS E6998-019, OpenShell: Architecture, Enforcement, and Policy (the course's guide to commit be7af99), chapter 1.

Each OpenShell component has its own job.[1]

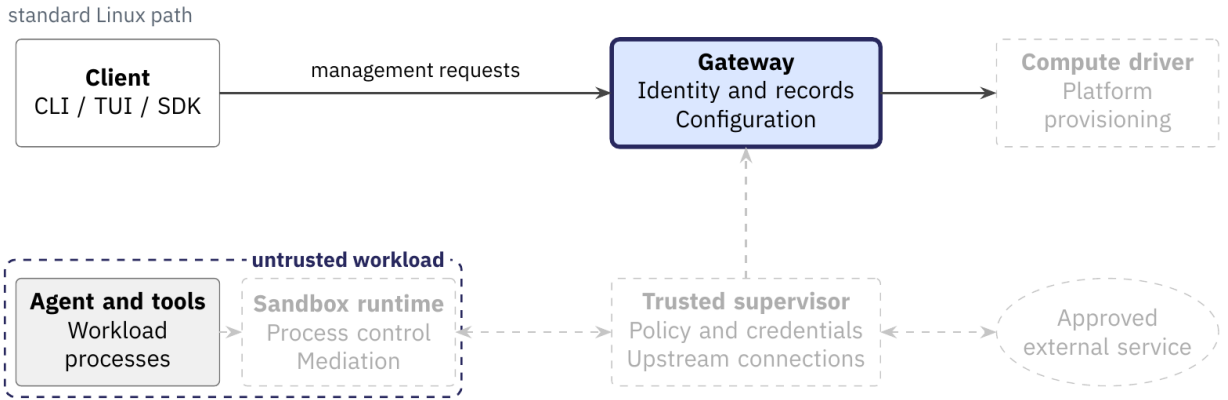
- **The client, gateway and driver** create, record and place sandboxes.



[1] COMS E6998-019, OpenShell: Architecture, Enforcement, and Policy (the course's guide to commit be7af99), chapters 2, 6 and 7.

Each OpenShell component has its own job.[1]

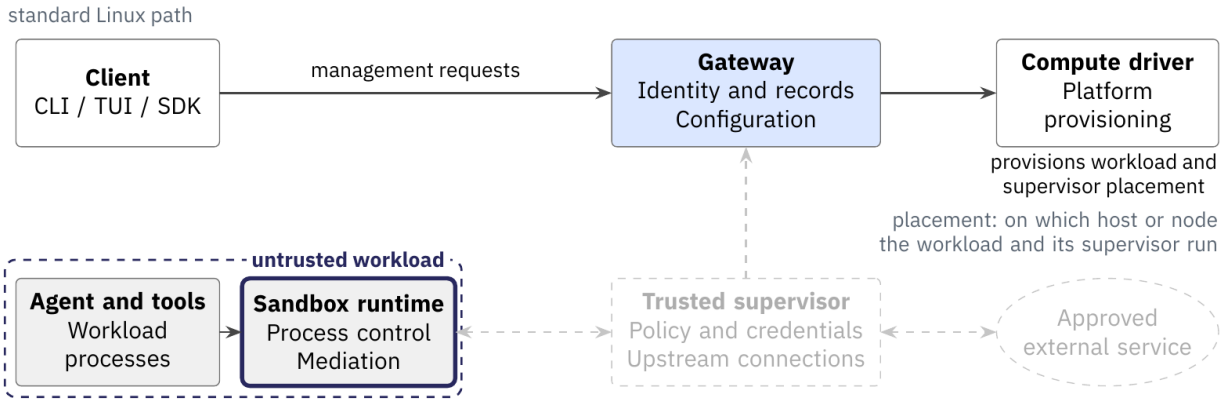
- **The client, gateway and driver** create, record and place sandboxes.



[1] COMS E6998-019, OpenShell: Architecture, Enforcement, and Policy (the course's guide to commit be7af99), chapters 2, 6 and 7.

Each OpenShell component has its own job.[1]

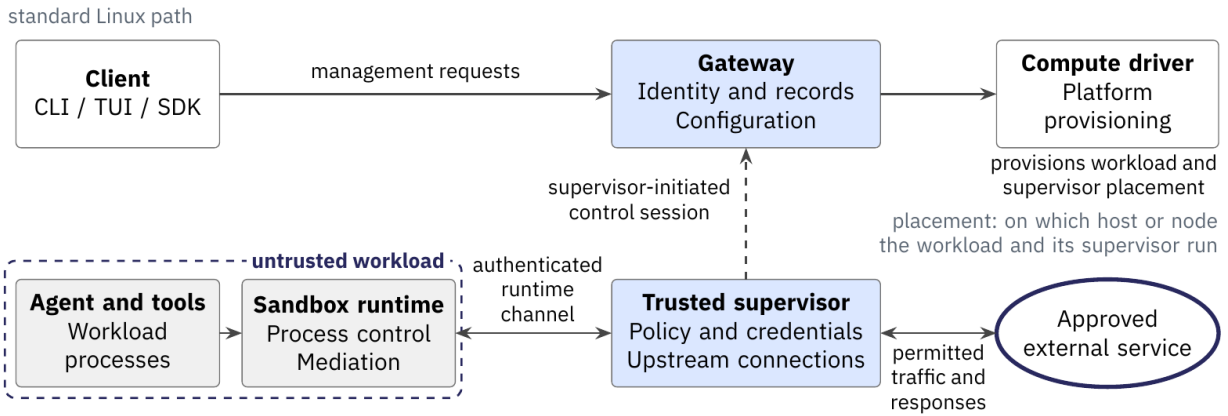
- **The client, gateway and driver** create, record and place sandboxes.
- **The sandbox runtime** fences the agent's processes from inside the workload.



[1] COMS E6998-019, OpenShell: Architecture, Enforcement, and Policy (the course's guide to commit be7af99), chapters 2, 6 and 7.

Each OpenShell component has its own job.[1]

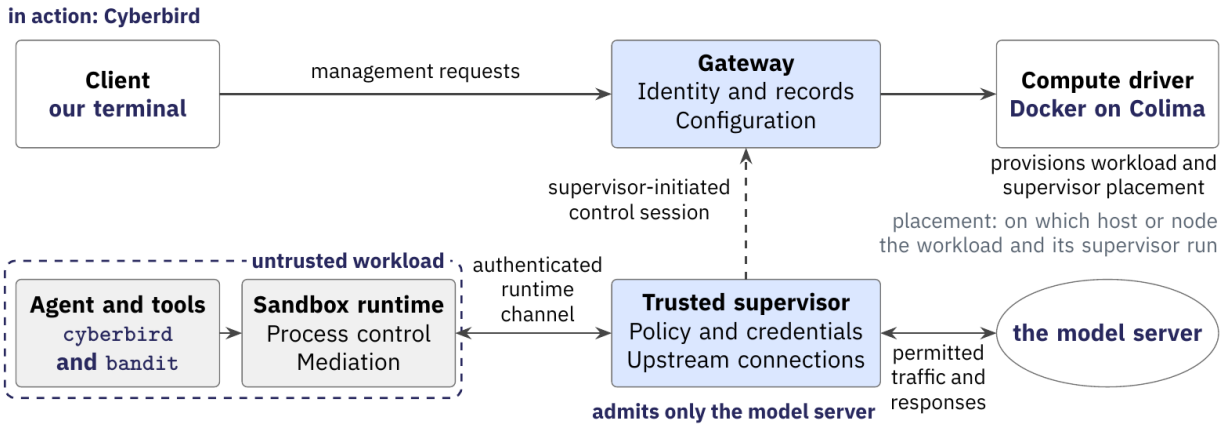
- **The client, gateway and driver** create, record and place sandboxes.
- **The sandbox runtime** fences the agent's processes from inside the workload.
- **The trusted supervisor** decides every connection from outside the workload.



[1] COMS E6998-019, OpenShell: Architecture, Enforcement, and Policy (the course's guide to commit be7af99), chapters 2, 6 and 7.

Each OpenShell component has its own job.[1]

- **The client, gateway and driver** create, record and place sandboxes.
- **The sandbox runtime** fences the agent's processes from inside the workload.
- **The trusted supervisor** decides every connection from outside the workload.
- **In Cyberbird's case,** Docker ran it, and only the model server was reachable.

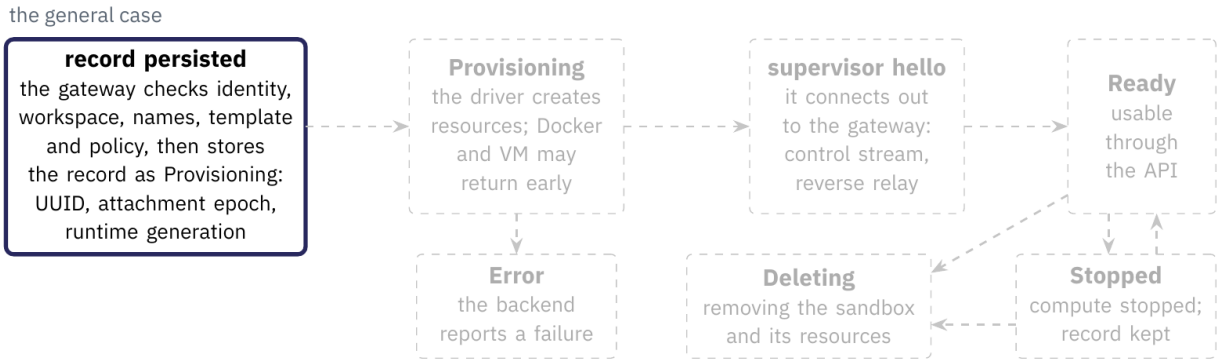


The keys and the network decisions stay outside the agent's workload.

[1] COMS E6998-019, OpenShell: Architecture, Enforcement, and Policy (the course's guide to commit be7af99), chapters 2, 6 and 7.

A sandbox is a record before a container.[1]

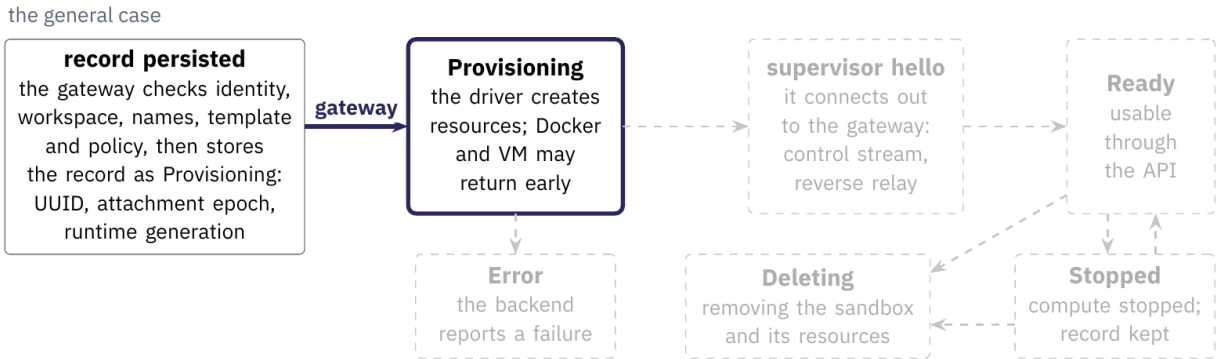
- **The record comes first:** the gateway stores it before any driver runs.



[1] COMS E6998-019, OpenShell: Architecture, Enforcement, and Policy (the course's guide to commit be7af99), chapter 4.

A sandbox is a record before a container.[1]

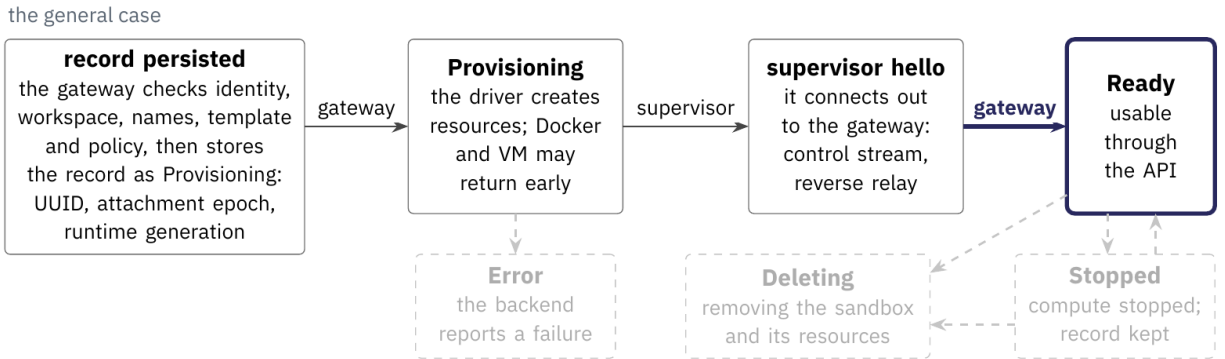
- **The record comes first:** the gateway stores it before any driver runs.
- **The driver provisions,** then the supervisor connects back and says hello.



[1] COMS E6998-019, OpenShell: Architecture, Enforcement, and Policy (the course's guide to commit be7af99), chapter 4.

A sandbox is a record before a container.[1]

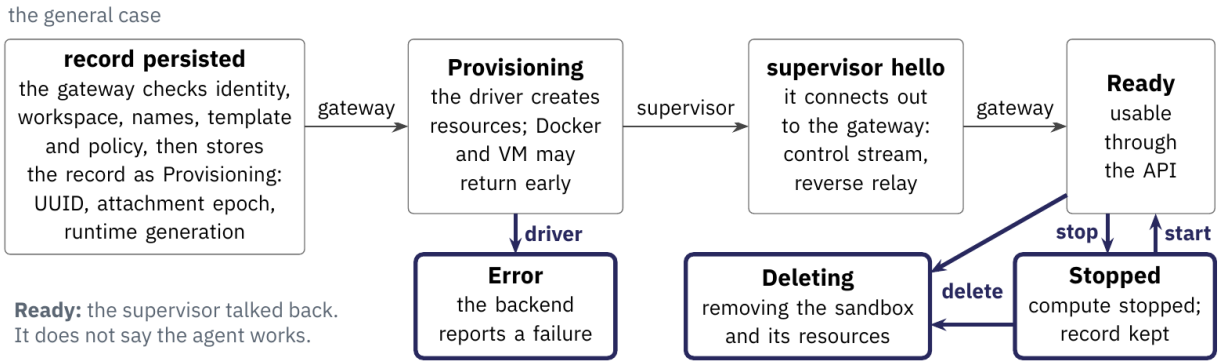
- **The record comes first:** the gateway stores it before any driver runs.
- **The driver provisions,** then the supervisor connects back and says hello.
- **Ready means the API can reach it,** and it says nothing about the agent.



[1] COMS E6998-019, OpenShell: Architecture, Enforcement, and Policy (the course's guide to commit be7af99), chapter 4.

A sandbox is a record before a container.[1]

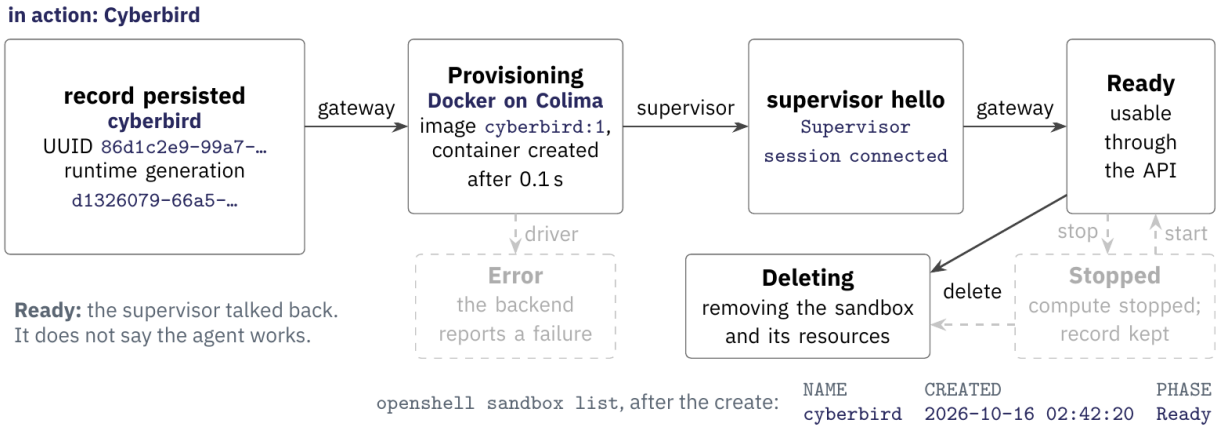
- **The record comes first:** the gateway stores it before any driver runs.
- **The driver provisions,** then the supervisor connects back and says hello.
- **Ready means the API can reach it,** and it says nothing about the agent.



[1] COMS E6998-019, OpenShell: Architecture, Enforcement, and Policy (the course's guide to commit be7af99), chapter 4.

A sandbox is a record before a container.[1]

- **The record comes first:** the gateway stores it before any driver runs.
- **The driver provisions,** then the supervisor connects back and says hello.
- **Ready means the API can reach it,** and it says nothing about the agent.
- **In Cyberbird's case,** `sandbox list` showed it Ready within four seconds.

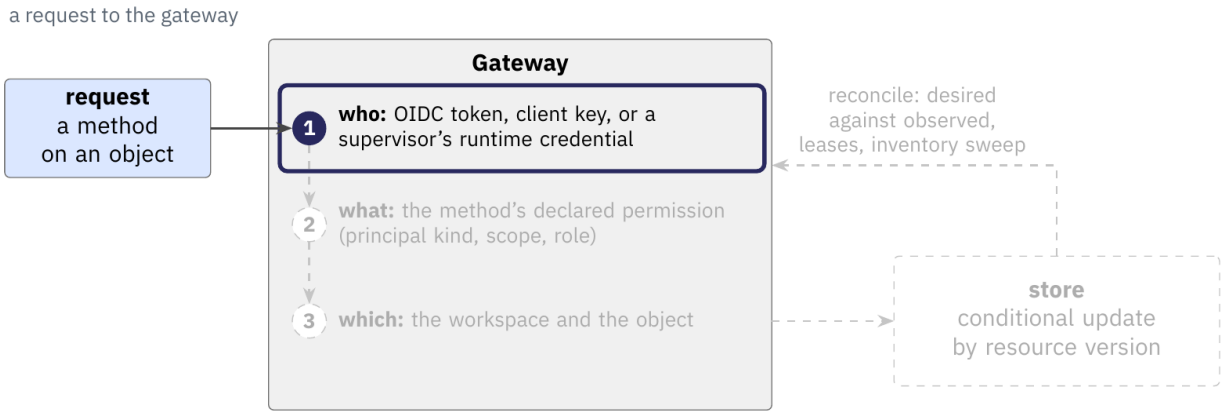


The gateway records each phase, so an interrupted create can be reconciled.

[1] COMS E6998-019, OpenShell: Architecture, Enforcement, and Policy (the course's guide to commit be7af99), chapter 4.

The gateway holds authority and records.[1]

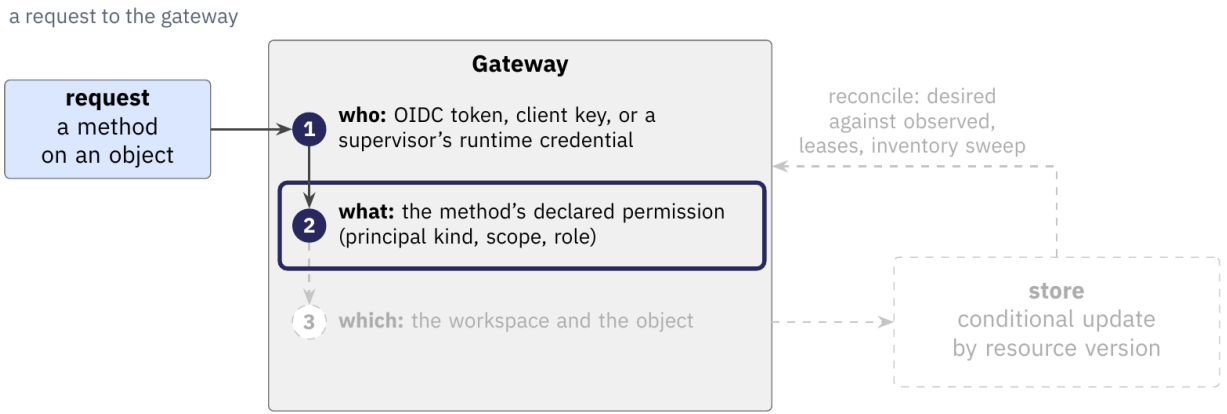
- **Who:** a user's token or client key, or a supervisor's own credential.



[1] COMS E6998-019, OpenShell: Architecture, Enforcement, and Policy (the course's guide to commit be7af99), chapter 5.

The gateway holds authority and records.[1]

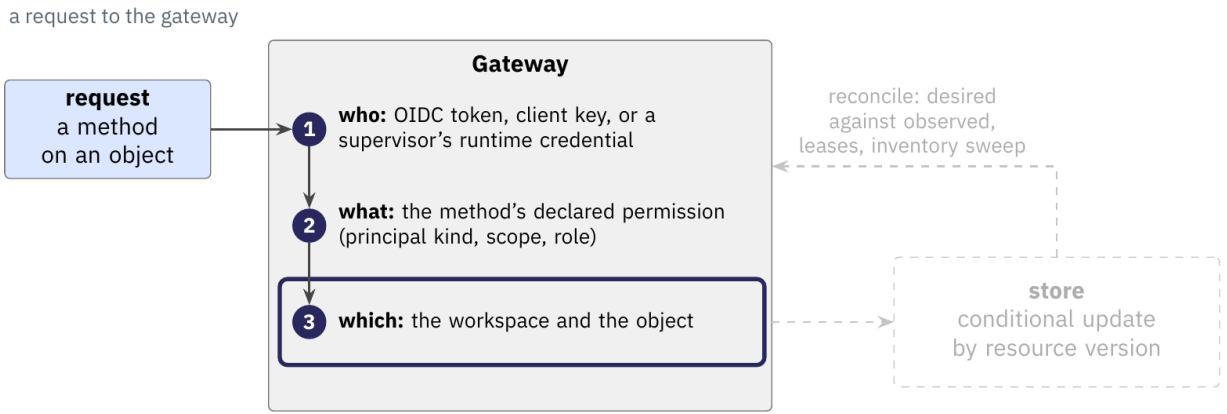
- **Who:** a user's token or client key, or a supervisor's own credential.
- **What and which:** the method's permission, then the workspace and object.



[1] COMS E6998-019, OpenShell: Architecture, Enforcement, and Policy (the course's guide to commit be7af99), chapter 5.

The gateway holds authority and records.[1]

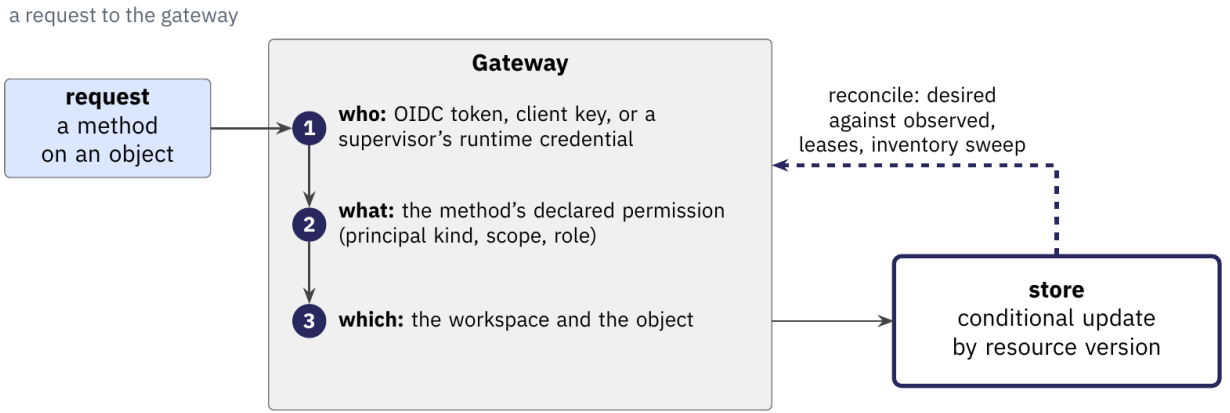
- **Who:** a user's token or client key, or a supervisor's own credential.
- **What and which:** the method's permission, then the workspace and object.



[1] COMS E6998-019, OpenShell: Architecture, Enforcement, and Policy (the course's guide to commit be7af99), chapter 5.

The gateway holds authority and records.[1]

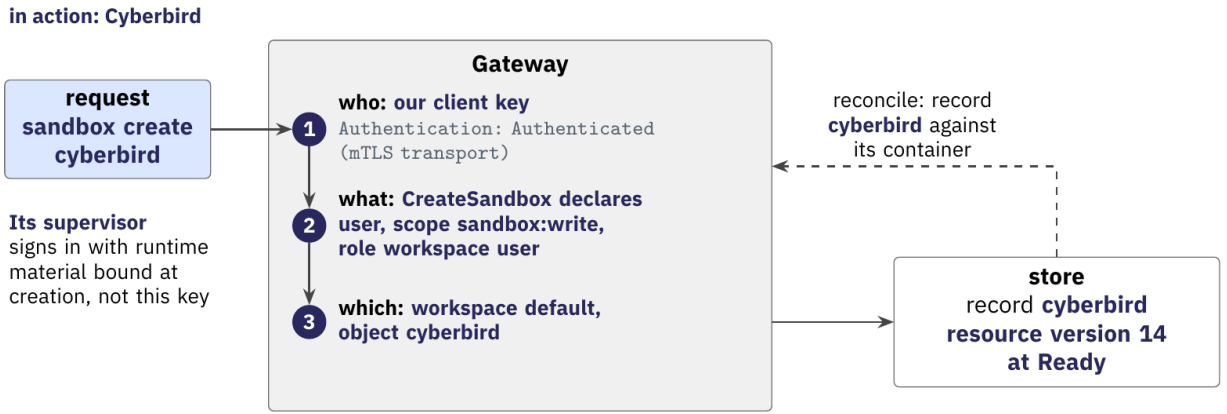
- **Who:** a user's token or client key, or a supervisor's own credential.
- **What and which:** the method's permission, then the workspace and object.
- **The store** writes a record only if its version is still the one read.



[1] COMS E6998-019, OpenShell: Architecture, Enforcement, and Policy (the course's guide to commit be7af99), chapter 5.

The gateway holds authority and records.[1]

- **Who:** a user's token or client key, or a supervisor's own credential.
- **What and which:** the method's permission, then the workspace and object.
- **The store** writes a record only if its version is still the one read.
- **In Cyberbird's case,** our client key created `cyberbird` in `default`.

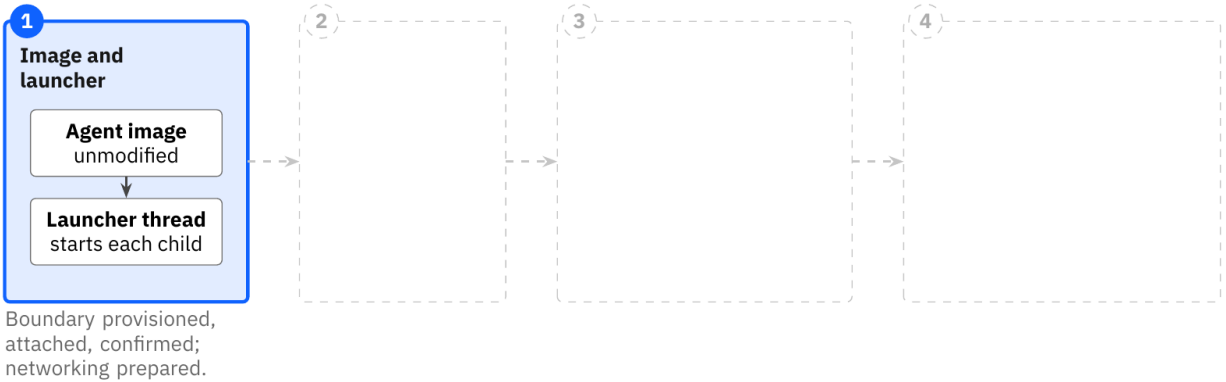


The gateway checks caller, method and object before a conditional write.

[1] COMS E6998-019, OpenShell: Architecture, Enforcement, and Policy (the course's guide to commit be7af99), chapter 5.

The agent starts with reduced privileges.[1]

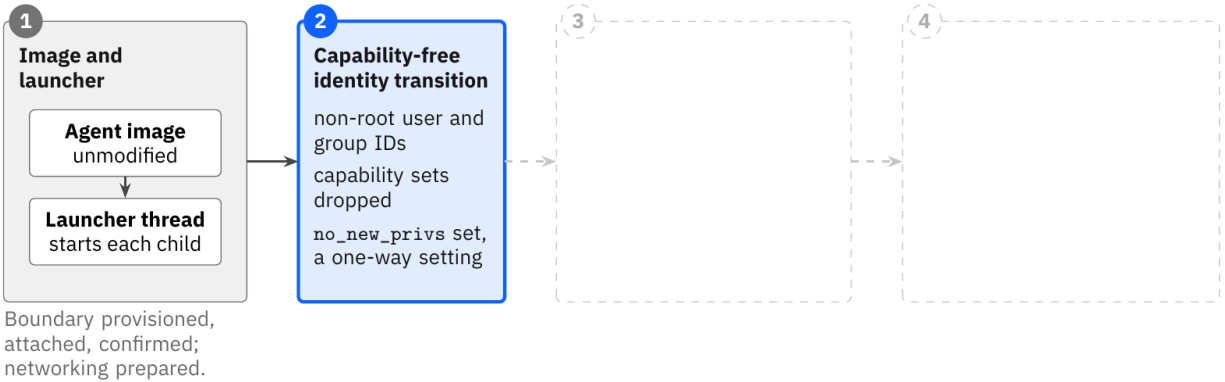
- **Each child drops privilege first:** non-root IDs and no capabilities.



[1] COMS E6998-019, OpenShell: Architecture, Enforcement, and Policy (the course's guide to commit be7af99), chapter 7.

The agent starts with reduced privileges.[1]

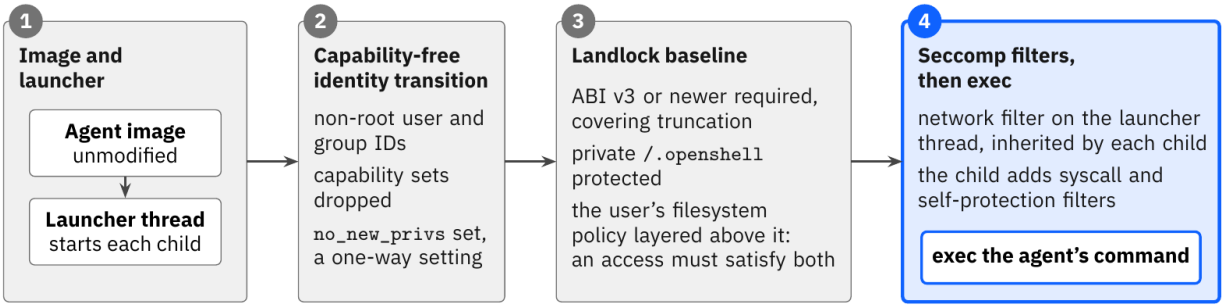
- **Each child drops privilege first:** non-root IDs and no capabilities.



[1] COMS E6998-019, OpenShell: Architecture, Enforcement, and Policy (the course's guide to commit be7af99), chapter 7.

The agent starts with reduced privileges.[1]

- **Each child drops privilege first:** non-root IDs and no capabilities.
- **Landlock** fences its files, above a baseline that guards the runtime.
- **Seccomp** blocks risky calls and pauses `connect()` for the broker.
- **In Cyberbird's case,** its user, uid 1000, could not list its own home directory.

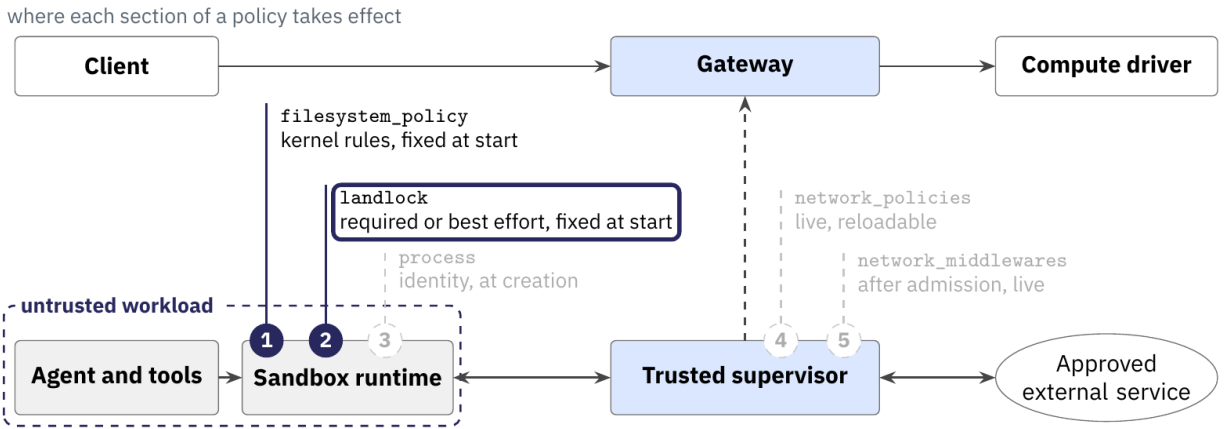


The fences are in the kernel before the agent's first instruction runs.

[1] COMS E6998-019, OpenShell: Architecture, Enforcement, and Policy (the course's guide to commit be7af99), chapter 7.

A policy states a workload's authority.[1]

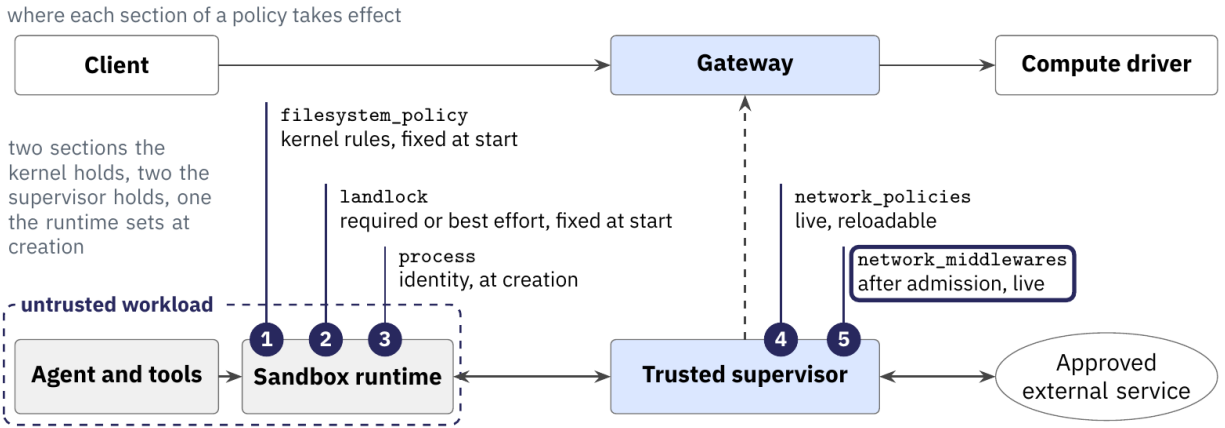
- **The kernel holds** `filesystem_policy` and `landlock`, fixed at the start.



[1] COMS E6998-019, OpenShell: Architecture, Enforcement, and Policy (the course's guide to commit be7af99), chapter 10.

A policy states a workload's authority.[1]

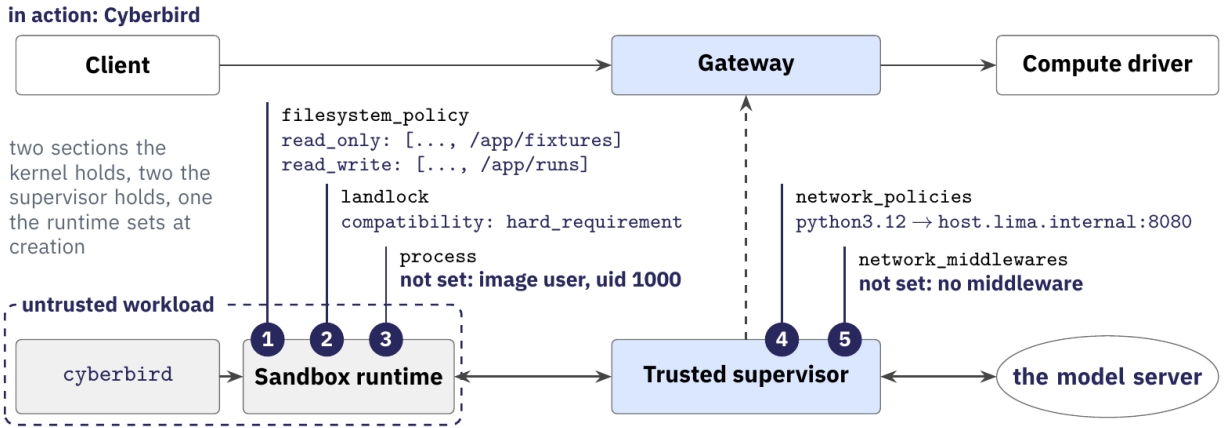
- **The kernel holds** `filesystem_policy` and `landlock`, fixed at the start.
- **The runtime sets** `process`, the workload's user and group, at creation.
- **The supervisor holds** `network_policies` and `network_middlewares`.



[1] COMS E6998-019, OpenShell: Architecture, Enforcement, and Policy (the course's guide to commit be7af99), chapter 10.

A policy states a workload's authority.[1]

- **The kernel holds** `filesystem_policy` and `landlock`, fixed at the start.
- **The runtime sets** `process`, the workload's user and group, at creation.
- **The supervisor holds** `network_policies` and `network_middlewares`.
- **In Cyberbird's case,** one rule lets `python3.12` reach the model server.



File rules are fixed at the start, and network rules can change live.

[1] COMS E6998-019, OpenShell: Architecture, Enforcement, and Policy (the course's guide to commit be7af99), chapter 10.

Cyberbird's policy, line by line[1]

- **As a document,** a policy is parsed, validated, versioned and diffed.
- **The read list grants** `/app`**,** so every case and the answer key stay readable.
- **One network rule** lets `python3.12` open TCP to the model server's port.

cyberbird-policy.yaml: the policy of the recorded run, its three comment lines left out

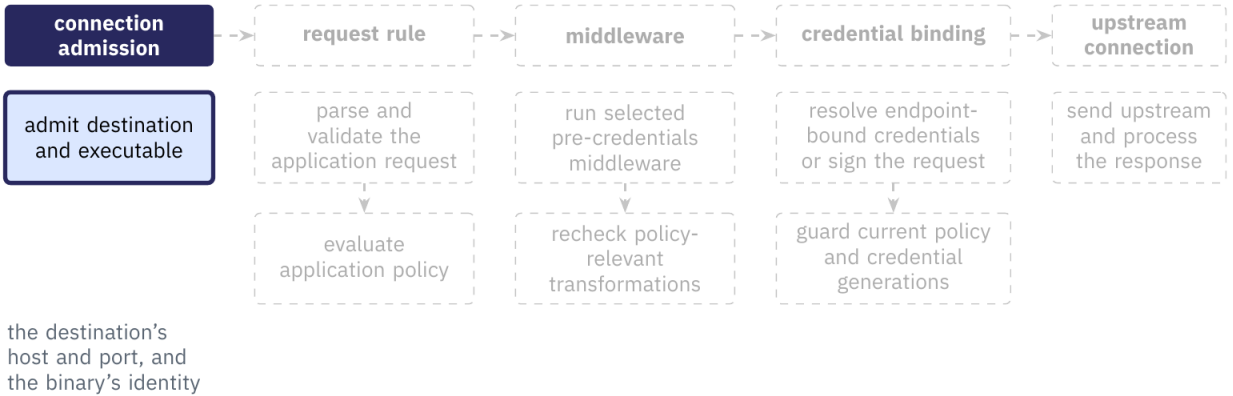
```yaml
version: 1
filesystem_policy:
  include_workdir: true
  read_only: [/bin, /usr, /lib, /etc, /proc, /dev/urandom, /app, /app/fixtures]
  read_write: [/sandbox, /tmp, /dev/null, /app/runs]
landlock:
  compatibility: hard_requirement
network_policies:
  model_server:
    endpoints:
      - host: host.lima.internal
        port: 8080
        protocol: tcp
    binaries:
      - path: /usr/local/bin/python3.12
```

Each grant covers everything beneath its path, the answer key included.

[1] COMS E6998-019, OpenShell: Architecture, Enforcement, and Policy (the course's guide to commit be7af99), chapter 10.

The supervisor checks connections first.[1]

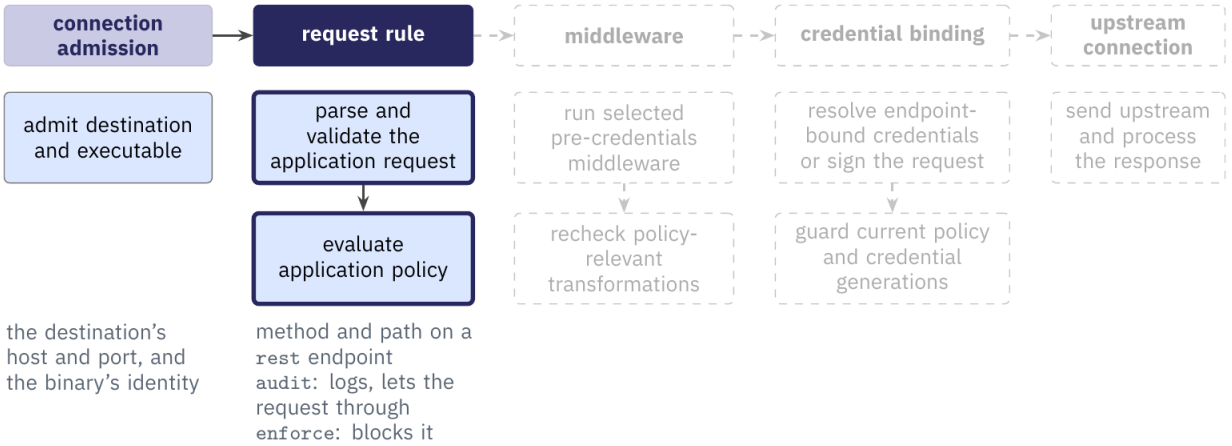
- **Connection admission** checks the host, the port and the calling binary.



[1] COMS E6998-019, OpenShell: Architecture, Enforcement, and Policy (the course's guide to commit be7af99), chapters 9 and 10.

The supervisor checks connections first.[1]

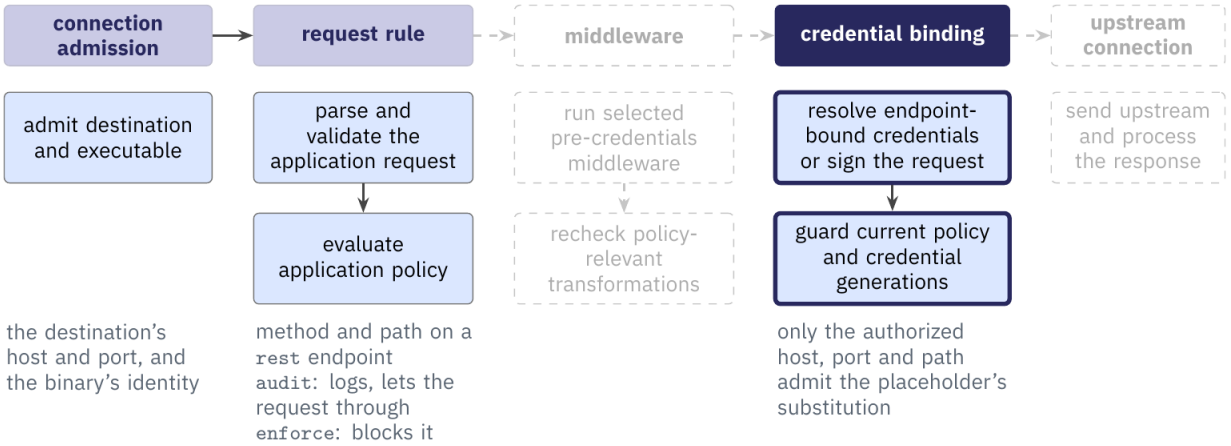
- **Connection admission** checks the host, the port and the calling binary.
- **Each request** then meets its rule, its credential binding and middleware.



[1] COMS E6998-019, OpenShell: Architecture, Enforcement, and Policy (the course's guide to commit be7af99), chapters 9 and 10.

The supervisor checks connections first.[1]

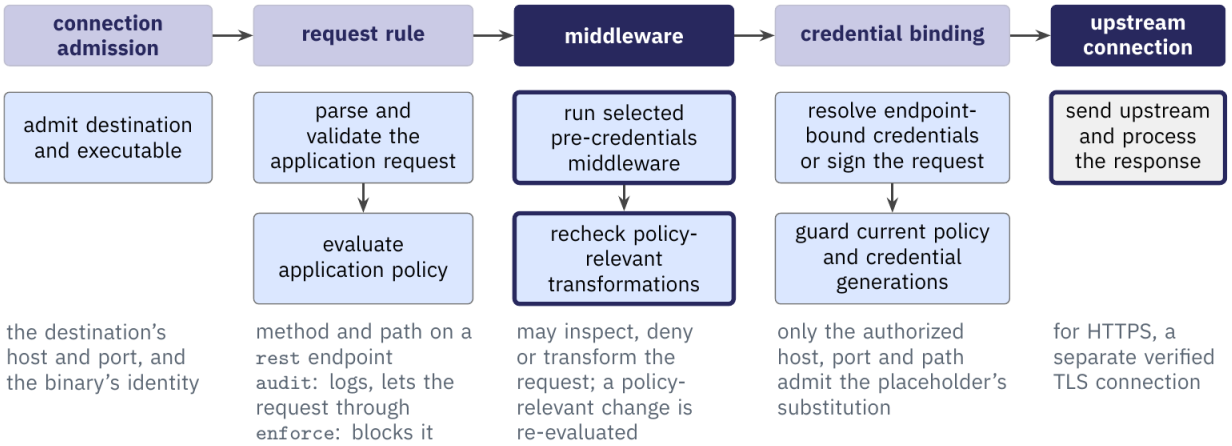
- **Connection admission** checks the host, the port and the calling binary.
- **Each request** then meets its rule, its credential binding and middleware.



[1] COMS E6998-019, OpenShell: Architecture, Enforcement, and Policy (the course's guide to commit be7af99), chapters 9 and 10.

The supervisor checks connections first.[1]

- **Connection admission** checks the host, the port and the calling binary.
- **Each request** then meets its rule, its credential binding and middleware.
- **In Cyberbird's case,** `python3.12` passed admission, and `pypi.org` failed it.

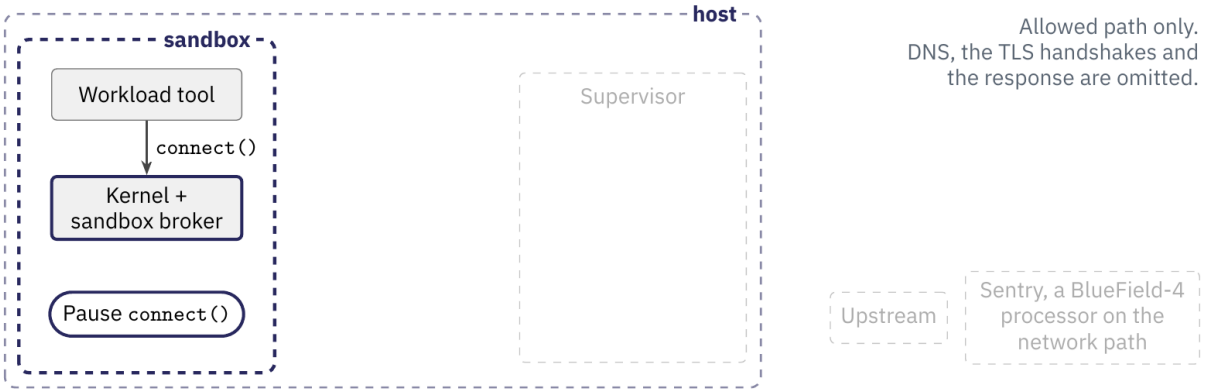


Audit, the default, logs a request-rule violation and lets the request through.

[1] COMS E6998-019, OpenShell: Architecture, Enforcement, and Policy (the course's guide to commit be7af99), chapters 9 and 10.

Every connection passes the supervisor.[1]

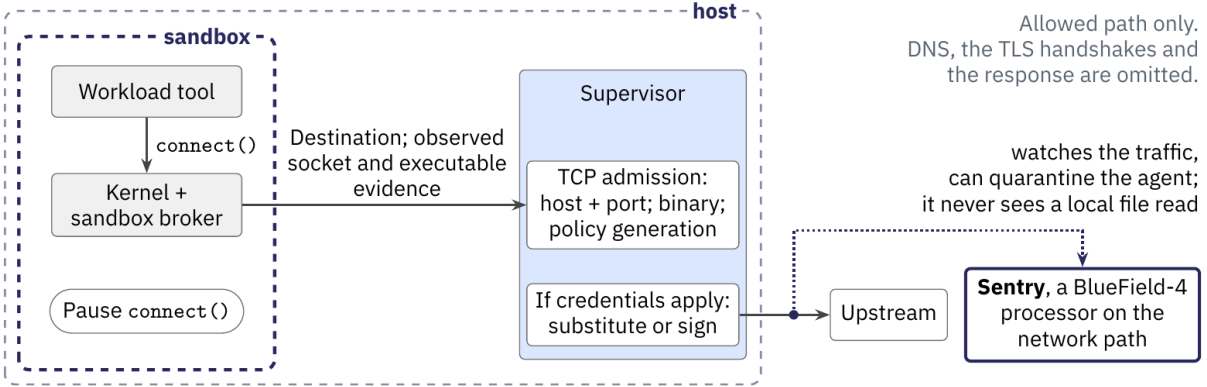
- **The kernel pauses** the program's `connect()` and hands it to the broker.



[1] COMS E6998-019, OpenShell: Architecture, Enforcement, and Policy (the course's guide to commit be7af99), chapters 2 and 8. [2] NVIDIA. Open Agent Safety Platform announcement, September 28, 2026. NVIDIA, 2026.

Every connection passes the supervisor.[1]

- **The kernel pauses** the program's `connect()` and hands it to the broker.
- **The supervisor admits it** by host, port, binary and policy generation.
- **The supervisor connects upstream** after substituting any credential.
- **In Cyberbird's case,** eight model calls took this path with no key to substitute.



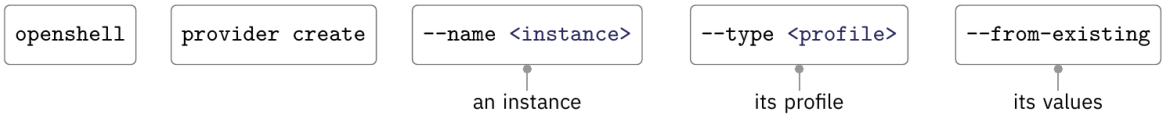
Each hop can stop a request, so each hop is a place to diagnose.

[1] COMS E6998-019, OpenShell: Architecture, Enforcement, and Policy (the course's guide to commit be7af99), chapters 2 and 8. [2] NVIDIA. Open Agent Safety Platform announcement, September 28, 2026. NVIDIA, 2026.

A provider keeps the key at the gateway.[1]

- **A provider instance** holds one account's credential in the gateway's store.

the shape of the command



[1] COMS E6998-019, OpenShell: Architecture, Enforcement, and Policy (the course's guide to commit be7af99), chapter 11. [2] NVIDIA. OpenShell, commit be7af99. https://github.com/NVIDIA/OpenShell

A provider keeps the key at the gateway.[1]

- **A provider instance** holds one account's credential in the gateway's store.

a real command, token by token

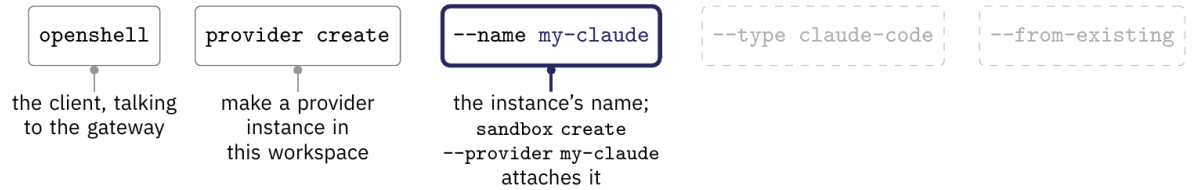


[1] COMS E6998-019, OpenShell: Architecture, Enforcement, and Policy (the course's guide to commit be7af99), chapter 11. [2] NVIDIA. OpenShell, commit be7af99. https://github.com/NVIDIA/OpenShell

A provider keeps the key at the gateway.[1]

- **A provider instance** holds one account's credential in the gateway's store.
- **Its name** is what `sandbox create --provider` attaches to one sandbox.

a real command, token by token

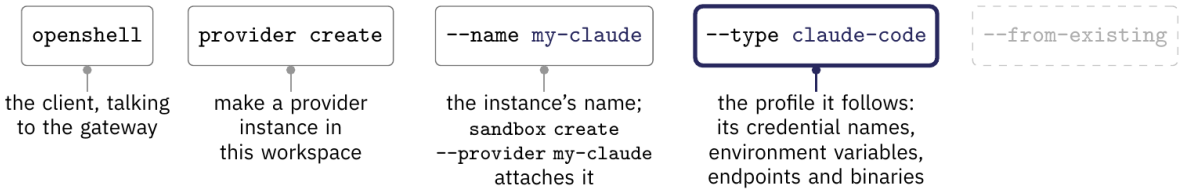


[1] COMS E6998-019, OpenShell: Architecture, Enforcement, and Policy (the course's guide to commit be7af99), chapter 11. [2] NVIDIA. OpenShell, commit be7af99. https://github.com/NVIDIA/OpenShell

A provider keeps the key at the gateway.[1]

- **A provider instance** holds one account's credential in the gateway's store.
- **Its name** is what `sandbox create --provider` attaches to one sandbox.
- **Its profile** names the key's variables, endpoints and permitted binaries.

a real command, token by token

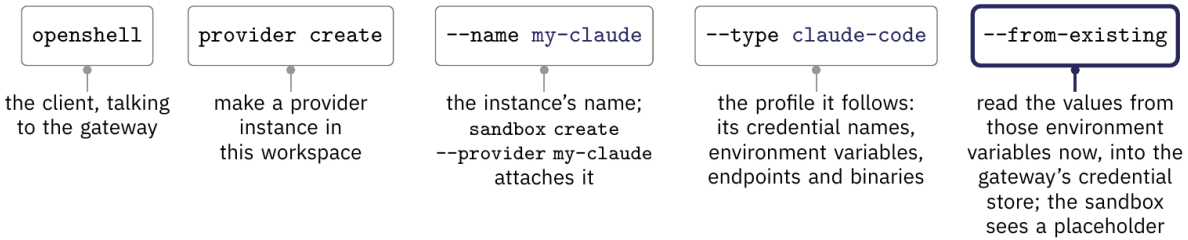


[1] COMS E6998-019, OpenShell: Architecture, Enforcement, and Policy (the course's guide to commit be7af99), chapter 11. [2] NVIDIA. OpenShell, commit be7af99. https://github.com/NVIDIA/OpenShell

A provider keeps the key at the gateway.[1]

- **A provider instance** holds one account's credential in the gateway's store.
- **Its name** is what `sandbox create --provider` attaches to one sandbox.
- **Its profile** names the key's variables, endpoints and permitted binaries.
- **In Cyberbird's case,** the profile must name `python3.12`, as in HW2.

in action: Cyberbird



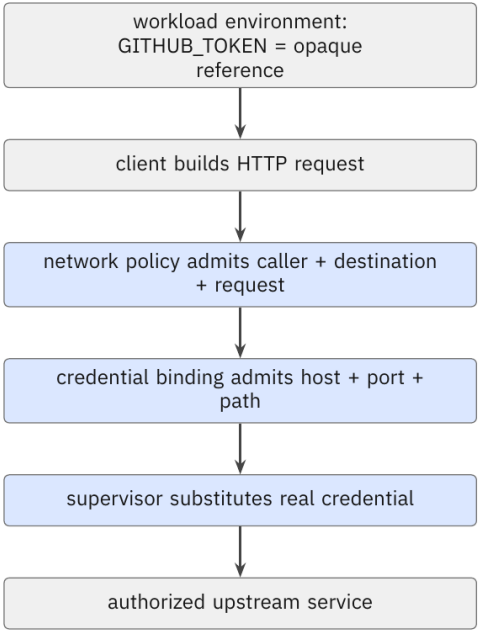
> Cyberbird with `--model anthropic:claude-opus-5` and `--provider my-claude` sees `ANTHROPIC_API_KEY` as a placeholder. In a test sandbox with a provider attached, `printenv ANTHROPIC_API_KEY | cut -c1-12` printed `openshell:re`.

The sandbox sees a placeholder, and the supervisor holds the real key.

[1] COMS E6998-019, OpenShell: Architecture, Enforcement, and Policy (the course's guide to commit be7af99), chapter 11. [2] NVIDIA. OpenShell, commit be7af99. https://github.com/NVIDIA/OpenShell

Real credentials never enter the sandbox.

- **Providers are named credential bundles** configured at the gateway and attached when a sandbox is created: `openshell provider create --name my-claude --type claude-code --from-existing`.
- **The agent receives a placeholder.** When a request leaves the sandbox for an authorized endpoint, the supervisor substitutes the real credential. Requests to other endpoints never see it.
- **Consequence:** credential theft from configuration finds nothing to read, and exfiltration through a tool call has no key to send.



Lecture 4's dispatcher supplied the workspace root because the model could not be trusted to. The supervisor supplies the key for the same reason.

A denied request becomes a proposal.[1]

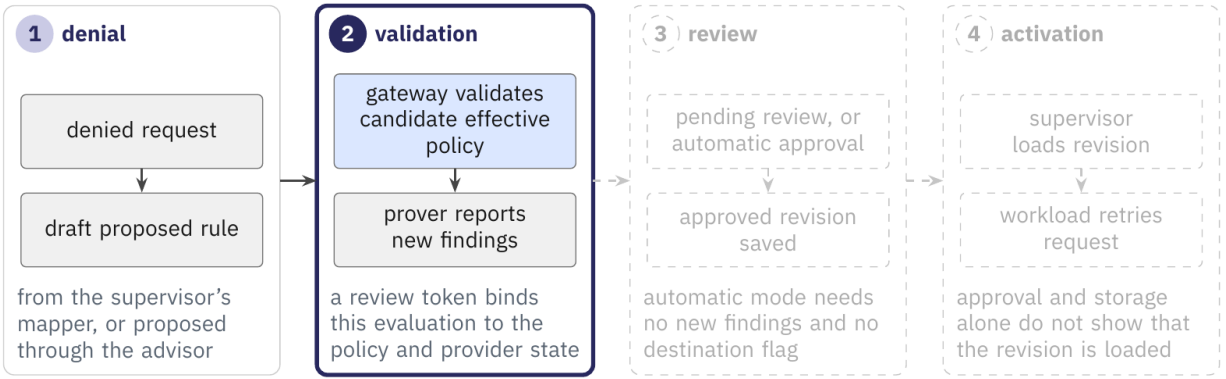
- **The denial is recorded,** and the mapper drafts a narrow rule from it.



[1] COMS E6998-019, OpenShell: Architecture, Enforcement, and Policy (the course's guide to commit be7af99), chapter 12.

A denied request becomes a proposal.[1]

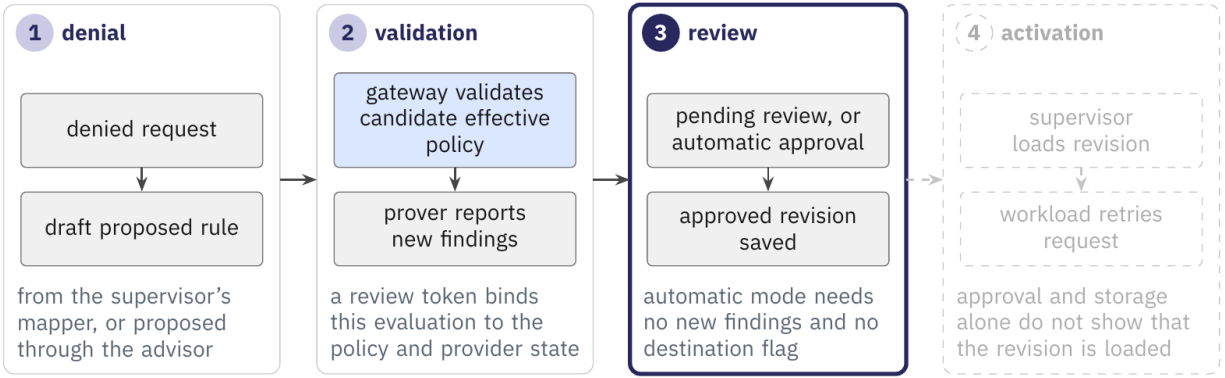
- **The denial is recorded,** and the mapper drafts a narrow rule from it.
- **The gateway validates** the candidate, and the prover reports new findings.



[1] COMS E6998-019, OpenShell: Architecture, Enforcement, and Policy (the course's guide to commit be7af99), chapter 12.

A denied request becomes a proposal.[1]

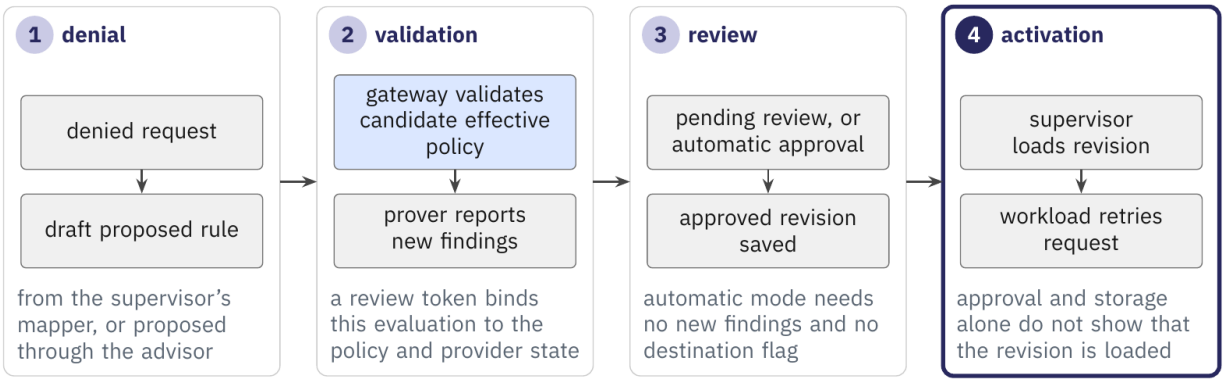
- **The denial is recorded,** and the mapper drafts a narrow rule from it.
- **The gateway validates** the candidate, and the prover reports new findings.
- **A reviewer approves it,** or automatic approval does when nothing is flagged.



[1] COMS E6998-019, OpenShell: Architecture, Enforcement, and Policy (the course's guide to commit be7af99), chapter 12.

A denied request becomes a proposal.[1]

- **The denial is recorded,** and the mapper drafts a narrow rule from it.
- **The gateway validates** the candidate, and the prover reports new findings.
- **A reviewer approves it,** or automatic approval does when nothing is flagged.
- **In Cyberbird's case,** a denied `pip` left a pending rule for `pypi.org:443`.



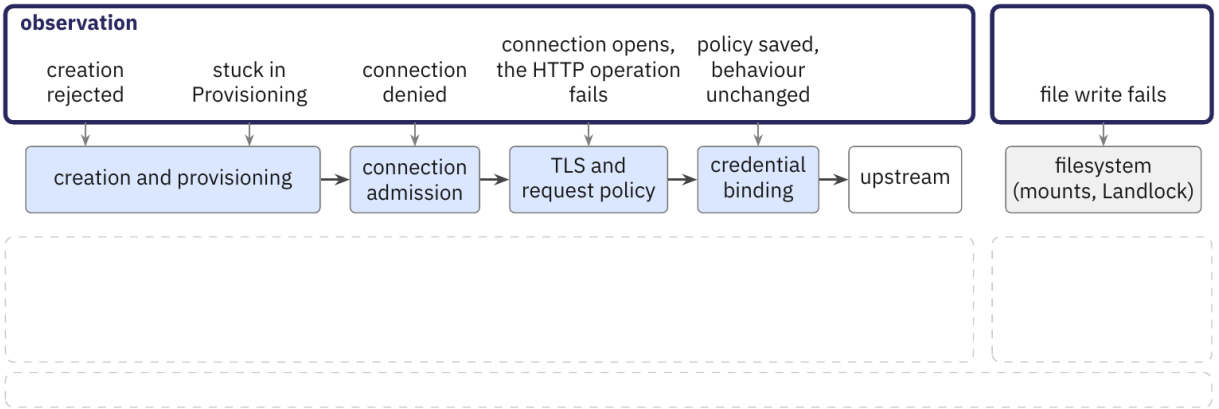
A proposal adds network rules and leaves file and identity rules alone.

[1] COMS E6998-019, OpenShell: Architecture, Enforcement, and Policy (the course's guide to commit be7af99), chapter 12.

Diagnose by the boundary that failed.[1]

A failure shows an unresolved condition at one boundary.

- **A request can stop** at any boundary, from its creation to the filesystem.

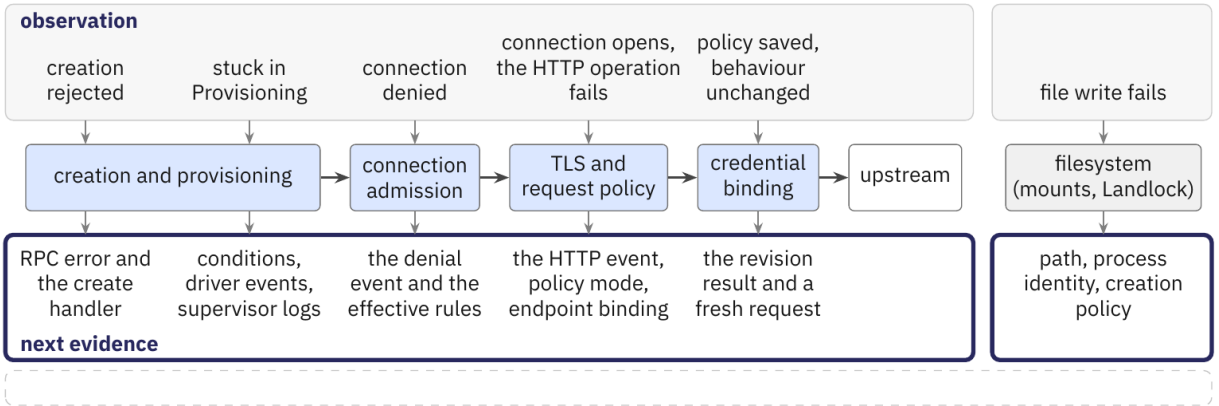


[1] COMS E6998-019, OpenShell: Architecture, Enforcement, and Policy (the course's guide to commit be7af99), chapter 14.

Diagnose by the boundary that failed.[1]

A failure shows an unresolved condition at one boundary.

- **A request can stop** at any boundary, from its creation to the filesystem.
- **Each boundary records** its own evidence: an error, an event or a result.

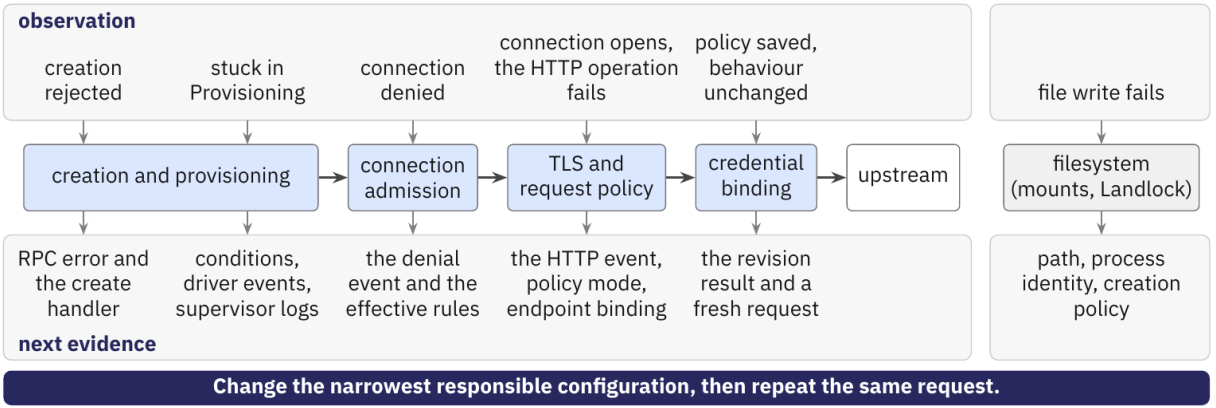


[1] COMS E6998-019, OpenShell: Architecture, Enforcement, and Policy (the course's guide to commit be7af99), chapter 14.

Diagnose by the boundary that failed.[1]

A failure shows an unresolved condition at one boundary.

- **A request can stop** at any boundary, from its creation to the filesystem.
- **Each boundary records** its own evidence: an error, an event or a result.
- **In Cyberbird's case,** `Connection error.` was a denial the log named.

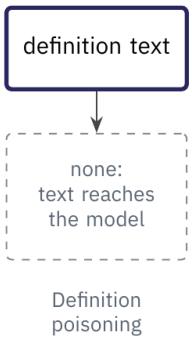


Diagnosis starts from the evidence of the boundary that failed.

[1] COMS E6998-019, OpenShell: Architecture, Enforcement, and Policy (the course's guide to commit be7af99), chapter 14.

What OpenShell does not cover.

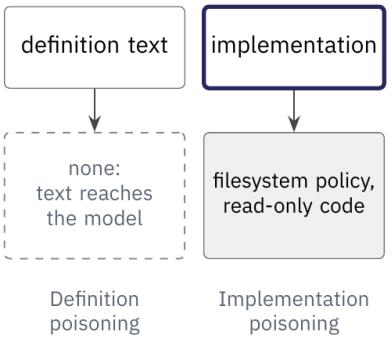
- **A tool's definition and response** reach the model as unchecked text[1].



[1] COMS E6998-019, OpenShell: Architecture, Enforcement, and Policy (the course's guide to commit be7af99), chapter 1. [2] Tigera. NVIDIA OpenShell Secures the Agent. Who Governs the Fleet? 2026 (a vendor's reading of OpenShell's documentation). [3] NVIDIA. Open Agent Safety Platform announcement, September 28, 2026. NVIDIA, 2026.

What OpenShell does not cover.

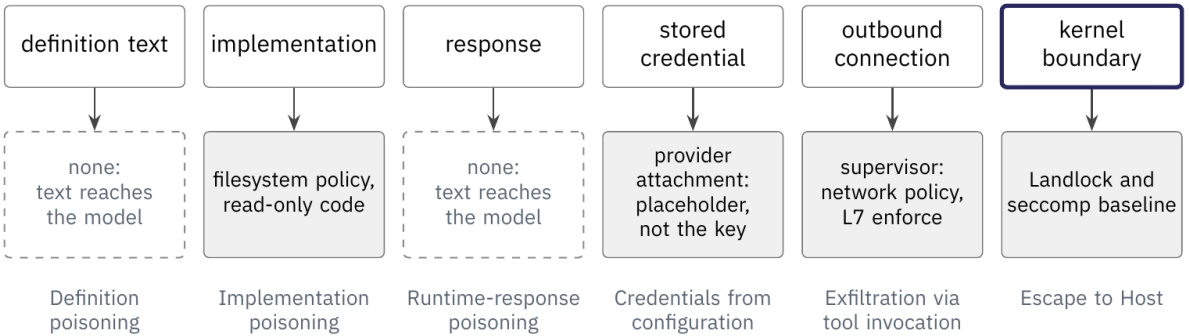
- **A tool's definition and response** reach the model as unchecked text[1].



[1] COMS E6998-019, OpenShell: Architecture, Enforcement, and Policy (the course's guide to commit be7af99), chapter 1. [2] Tigera. NVIDIA OpenShell Secures the Agent. Who Governs the Fleet? 2026 (a vendor's reading of OpenShell's documentation). [3] NVIDIA. Open Agent Safety Platform announcement, September 28, 2026. NVIDIA, 2026.

What OpenShell does not cover.

- **A tool's definition and response** reach the model as unchecked text[1].
- **The agent holds a placeholder** and has no identity of its own.
- **Governing agents across sandboxes** lies outside its scope[2].
- **Sentry**[3]**,** the watcher outside the host, runs on hardware the course lacks.
- **In Cyberbird's case,** the model believes a case file OpenShell never reads.

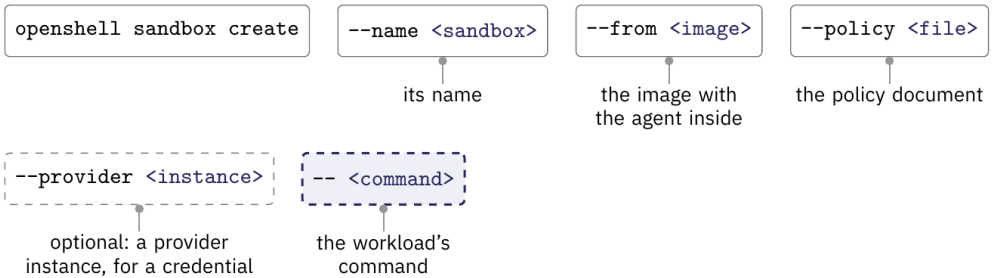


The host must check what the model reads, since OpenShell cannot.

[1] COMS E6998-019, OpenShell: Architecture, Enforcement, and Policy (the course's guide to commit be7af99), chapter 1. [2] Tigera. NVIDIA OpenShell Secures the Agent. Who Governs the Fleet? 2026 (a vendor's reading of OpenShell's documentation). [3] NVIDIA. Open Agent Safety Platform announcement, September 28, 2026. NVIDIA, 2026.

The commands HW2 uses.

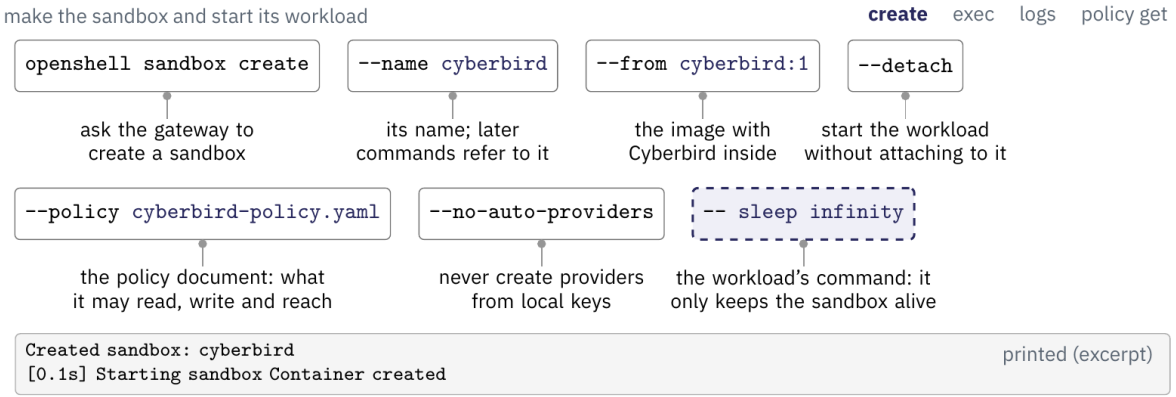
the shape of `sandbox create`



Before the double dash, the operator's flags, which OpenShell reads. After it, the workload's command, which runs inside the sandbox.

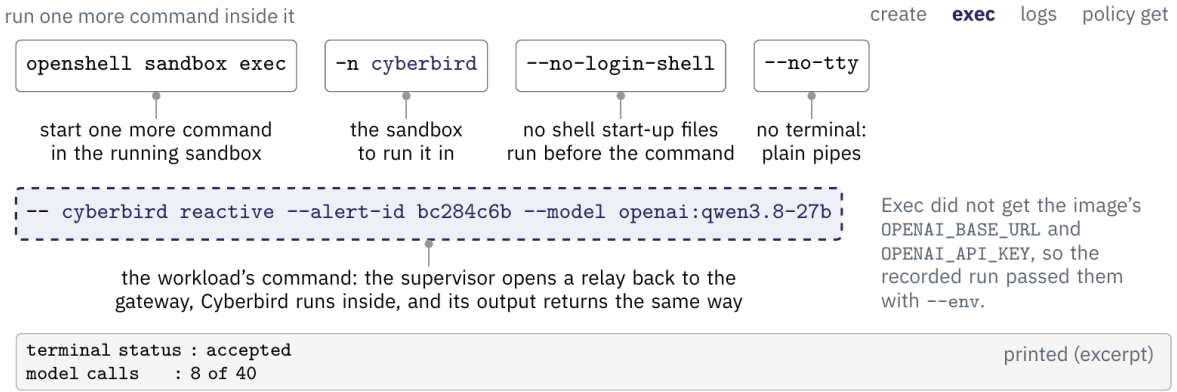
The commands HW2 uses.

- **Create:** `--from` names the image and `--policy` the policy file.



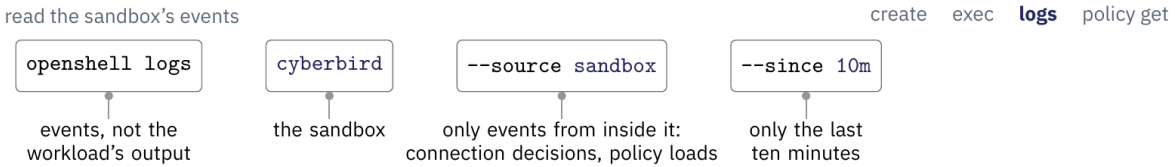
The commands HW2 uses.

- **Create:** `--from` names the image and `--policy` the policy file.
- **Exec:** `--no-login-shell --no-tty` keep an MCP stream clean.



The commands HW2 uses.

- **Create:** `--from` names the image and `--policy` the policy file.
- **Exec:** `--no-login-shell --no-tty` keep an MCP stream clean.
- **Logs:** `--source sandbox` shows its denials and policy loads.



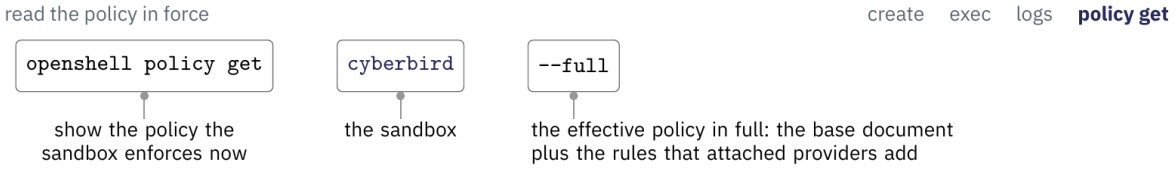
printed (excerpt)

```
NET:OPEN [MED] DENIED /usr/local/bin/python3.12(0) -> pypi.org:443 ...
NET:OPEN [MED] DENIED /usr/bin/curl(0) -> host.lima.internal:8080 ...
NET:OPEN [INFO] ALLOWED /usr/local/bin/python3.12(0) -> host.lima.internal:8080 ...
```

The policy admits a binary together with a destination: python3.12 may reach the model server, curl may not.

The commands HW2 uses.

- **Create:** `--from` names the image and `--policy` the policy file.
- **Exec:** `--no-login-shell --no-tty` keep an MCP stream clean.
- **Logs:** `--source sandbox` shows its denials and policy loads.
- **Policy:** `--full` prints the effective policy, provider rules included.



printed (excerpt)

```
Status:     Effective
Source:     sandbox
...
landlock:
  compatibility: hard_requirement
network_policies:
  model_server:
    ...
```

No provider is attached to this sandbox, so the effective policy holds no provider rules.

Flags before the double dash are the operator's. The sandbox list then read: cyberbird 2026-10-16 02:42:20 Ready.

Models escape containers with known flaws.

These four slides summarize research on how kernel and container mechanisms fail and how their authority is revoked.

- **SandboxEscapeBench (ICML 2026)** measures whether frontier models can exploit seeded vulnerabilities to escape OCI containers running inside a virtual machine: misconfiguration, privilege mistakes, kernel flaws, orchestration weaknesses. When vulnerabilities are present, the models find and exploit them[1].
- **The benchmark covers Docker and OCI containers and Kubernetes pods only.** It does not cover gVisor, Firecracker or libkrun.
- **Across five sandbox products (Andronchik and Lokhmakov, preprint, 2026),** engine-side patches for coordinated disclosures land in about zero days, while the lag before products adopt them spans 0 to 471 or more days[2].

Models exploit seeded container flaws, and products adopt engine patches after a lag of 0 to 471 or more days.

CS @CU

[1] Quantifying Frontier LLM Capabilities for Container Sandbox Escape (SandboxEscapeBench). arXiv:2603.02277, 2026. [2] AI Code Sandboxes: A Comparative Security Study, Part 1. arXiv:2606.08433, 2026.

Sandlock writes the threat model down.

- **Sandlock (workshop preprint, May 2026)** compiles static, input-independent policy into Landlock and seccomp-bpf, and runs a narrow supervisor over seccomp user notification for decisions that depend on run-time state[1].
- **It adds a programmable run-time policy function, HTTP-level access control, and reversible filesystem effects,** at about 5 ms of startup overhead and bare-metal Redis throughput. It needs no root, no cgroups, no images. It is open source.
- **Trusted:** the host kernel, the supervisor and the embedding agent runtime. **Out of scope:** kernel vulnerabilities, side channels, privileged local attackers.

Sandlock builds on the kernel primitives behind OpenShell's locked layers and states its assumptions.

CS @CU

[1] Wang and Zheng. Sandlock: Confining AI Agent Code with Unprivileged Linux Primitives. Agentic OS Workshop at ASPLOS 2026, arXiv:2605.26298, 2026.

Research asks how authority is taken back.

- **Lingering Authority (Santos-Grueiro, preprint, June 2026):** revocable capabilities for coding agents. Authority granted for one subgoal should not outlive it. Its prototype PORTICO rejects all 10 post-closure reuses that a non-revoking comparator permits[1].
- **ActPlane (Zheng et al., preprint, June 2026):** agents declare policies, and an eBPF engine enforces them in the kernel with semantic feedback, at 1.9% to 8.4% overhead, covering execution paths that tool-call interception cannot see[2].
- **AgentKernel (Zou et al., preprint, August 2026):** "current governance stacks remain application-level middleware that share a process trust boundary with the agents they monitor." Its answer is a mandatory boundary with kernel-level allowlists and taint-aware memory[3].
- **Wang, Li and Tian (preprint, May 2026):** a survey of 59 papers, 21 production agent systems and 26 security plugins finds that production scope mostly widens and rarely shrinks[4].

Stronger walls are the premise. The research question is how authority is granted and taken back.

CS @CU

[1] Santos-Grueiro. Lingering Authority. arXiv:2606.22504, 2026. [2] Zheng et al. ActPlane. arXiv:2606.25189, 2026. [3] Zou et al. AgentKernel. arXiv:2609.29647, 2026. [4] Wang, Li and Tian. Reframing LLM Agent Security as an Agent-Human Interaction Problem. arXiv:2605.24309, 2026.

Peer review trails deployment.

- **Andronchik and Lokhmakov (preprint, June 2026)** read six engine-level measurements across five products: attack surface, leakage, stackability, CVE history, patch cadence, fuzzing. Engine classes (microVM, user-space kernel, OCI container) separate cleanly. Products within a class do not. The title announces a Part 2[1].
- **MCP-SandboxScan (preprint, January 2026)** runs MCP tools under WebAssembly or drives them over stdio and reports source-to-sink witnesses: on a 100-repository corpus it profiles 1,127 tools in 71 repositories, 886 of them with security-sensitive declared capabilities, and observes witnesses in 12 of the 33 repositories it re-executes[2].
- **Open:** no peer-reviewed evaluation of OpenShell; the escape benchmark covers OCI containers only; no first-class agent identity; no graduated enforcement, which AgentKernel proposes.

OpenShell is deployed, and the research on the mechanisms beneath it is mostly 2026 preprints.

CS @CU

[1] AI Code Sandboxes: A Comparative Security Study, Part 1. arXiv:2606.08433, 2026. [2] MCP-SandboxScan (SandScope). arXiv:2601.01241, 2026.

The agent must not reach the runtime.

- **The server validates argument and result schemas.** Descriptions and effect annotations provide no enforcement. Execution constraints depend on checks implemented by the host and server.
- **ATLAS describes the adversary's action and the asset it acts on:** the definition text, the implementation, the response, the stored credential, the outbound connection, the kernel boundary.
- **The runtime determines the enforcement point for each asset:** the host for text, the kernel for code and files, the supervisor for connections and credentials.
- **A rule enforced inside the agent's process can be disabled by the agent.** A rule enforced outside it cannot be disabled without compromising the enforcing process.
- **A rule enforced only in the model's alignment weakened once eleven models were given tools,** with no attack on the tools (slide 38), so the model should not be the enforcement point for its own tool use.
- **Last week:** the guarantees are supplied by the contract and the runtime.

This week: the runtime must execute in a process the agent cannot modify. The model cannot serve as the runtime.

Where should the ~/.ssh rule be enforced?

The rule is: the agent may not read ~/.ssh. Where should the rule be enforced?

- **a** In the system prompt
- **b** In the harness's tool code
- **c** **In the operating-system kernel**
- **d** In hardware outside the host

The evidence supports (c): the kernel enforces the rule from outside the agent's process. A processor in the network path never sees a local file read.

What the midterm covers

- **Dynamic control** and who chooses the next operation.
- **State records, routing and termination;** token accounting and admission.
- **Tool contracts and the dispatch cycle;** action representation, structured and code actions.
- **MCP hosts, clients, servers** and transports.
- **The tool boundary under attack,** and where a rule is enforced.

One hour, individual, closed book. The questions ask for reasoning about scenarios rather than recall of APIs.

Next Friday is the midterm.

- **Friday, October 23:** the midterm, one hour, individual, closed book, units 1 and 2 as taught. Week 8, October 30, teaches concurrency and multi-agent patterns. HW1 is due tonight before 11:59 pm. HW2 is released today and due Friday, November 13.
- **Deferred to week 11:** failure semantics, including which component makes a side effect happen once.

A rule belongs where the adversary cannot reach it from inside the agent.